

Linear Descent for Rank-2 and Rank-4 Module-LIP

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Abstract. Module-LIP asks to recover an isometry between Hermitian module lattices that carry a public number-field action. This action exposes the ambient $\mathbb{L}$ -linear structure and therefore permits tensor constructions that are unavailable in ordinary LIP. The main difficulty is not merely to obtain a smaller lattice: a useful descent must also identify which short vectors encode the hidden transition and give an exact procedure for lifting them back to the original Module-LIP instance. We give such end-to-end linear descents for determinant-one instances in module ranks two and four, writing $d = [\mathbb{L}^+ : \mathbb{Q}]$. In rank two, a conjugate-linear symmetry on V induces a three-dimensional $\mathbb{L}^+$ -fixed space in $\text{Sym}_{\mathbb{L}}^2(V)$. Its integral points form a rank- $3d$ lattice whose shortest vectors are exactly the integral self-adjoint involutions. A matched source–target pair of involutions reveals two rank-one submodules; Lenstra–Silverberg recovery on these lines, followed by exact module verification, reconstructs the transition. For the standard power-of-two cyclotomic orbit, the source shortest shell is explicit, so a single γ -SVP output with $\gamma < \sqrt{2}$ suffices. Using the Bambury–Nguyen primal–dual reduction, all oracle calls can be restricted to ranks at most $\lfloor 3d/2 \rfloor + 1$. In rank four, the Hermitian Hodge involution on $\bigwedge_{\mathbb{L}}^2 V$ yields a rank- $6d$ auxiliary lattice. For an orthogonal pseudobasis, we describe this lattice as three weighted ideal lattices. Given a matched exterior-square map, four image lines are recovered by elementary wedge-annihilator computations, and four rank-one Lenstra–Silverberg calls reconstruct and verify the original transition. For the standard cyclotomic orbit, the Hodge lattice is hypercubic; its public $\mathcal{O}_{\mathbb{L}^+}$ -action reduces basis matching to a polynomial signed-cycle enumeration. Bambury–Nguyen then lowers the largest oracle rank to $3d + 1$. The fixed-lattice constructions and matched-data decoders hold more generally, while completing SVP output to a matched rank-four map beyond the standard hypercubic family remains open.

Keywords: Module-LIP · linear descent · symmetric square · Hodge star · exterior square · shortest vector problem

1 Introduction

The module lattice isomorphism problem (Module-LIP) asks for a module isometry between two promised-isometric Hermitian lattices equipped with a public action of a number-field order. In rank two over power-of-two cyclotomic rings,

this is the structured isomorphism problem that underlies HAWK [5,6]. Unlike in ordinary LIP, the order action is part of the public input. Together with the Euclidean Gram matrix, it recovers the ambient $\mathbb{L}$ -valued Hermitian form and makes tensor and exterior-power representations over $\mathbb{L}$ algorithmically available.

This extra structure suggests a natural attack: move the hidden isometry to a lower-dimensional representation and solve a lattice problem there. Yet a smaller lattice alone does not give a search reduction. Three ingredients must fit together:

1. the auxiliary lattice and its metric must be computable from the public input and transported by every hidden isometry;
2. the relevant SVP output must be recognizable inside that lattice; and
3. the recognized auxiliary data must determine a transition for the original Module-LIP instance, including its integral module structure.

The second and third requirements are easy to blur. In particular, an isometry of an auxiliary lattice need not lie in the tensor representation of an isometry of the original module, and independently recovered rank-one pieces need not glue integrally. Our main goal is to close this full chain before using any oracle-rank reduction.

Several algebraic attacks on rank-two Module-LIP are known under different hypotheses [14,13,7], with further variants in [3,1,8,9]. Luo, Jiang, Pan, and Wang use the rank-two symplectic identity to construct a conjugated pseudo-automorphism whose eigenspaces expose rank-one submodules [13]; the same recovery mechanism is used in the subsequent commitment work [12]. Independently, Bambury and Nguyen show that certain SVP and hypercubic-basis tasks can be performed with SVP oracles in essentially half the ambient rank [2]. We use the rank-one recovery and oracle reductions as black boxes. The contribution here is to identify the appropriate rank-two and rank-four auxiliary lattices, prove exact decoders back to Module-LIP, and determine when the auxiliary SVP problem can be completed to the matched data required by those decoders.

We work over a CM extension $\mathbb{L}/\mathbb{L}^+$ and write

$$R = \mathcal{O}_{\mathbb{L}}, \quad S = \mathcal{O}_{\mathbb{L}^+}, \quad d = [\mathbb{L}^+ : \mathbb{Q}], \quad V = \mathbb{L} \otimes_R P.$$

The main statements concern the normalized problem in which the two Hermitian determinants and the transition determinant are one. This normalization makes the relevant symplectic and exterior-power identities exact. For free $\mathrm{GL}_r(R)$ instances based at a determinant-one form, the Lenstra–Silverberg norm-recovery procedure reduces the problem, after finitely many root-of-unity branches, to the normalized setting; see [proposition 4](#).

1.1 Results and scope

Our results separate statements that hold for arbitrary projective modules from the additional geometry needed to obtain matched auxiliary data by SVP.

1. *Rank-two symmetric-square descent.* For every normalized rank-two projective module, we construct a public rank- $3d$ lattice in $\text{Sym}_{\mathbb{L}}^2(V)$. A natural tensor-to-matrix map identifies it with the integral trace-zero self-adjoint endomorphisms, and its unit shell consists exactly of involutions. Given a matched pair of source and target involutions, two Lenstra–Silverberg rank-one recoveries, followed by the original module and Hermitian tests, recover a valid transition. For the standard power-of-two cyclotomic orbit, the complete source shortest shell is explicit. Consequently one target γ -SVP solution with $\gamma < \sqrt{2}$ closes the direct reduction.
2. *Rank-four Hodge descent and elementary lifting.* For every normalized rank-four projective module, the Hermitian Hodge involution gives a public rank- $6d$ lattice in $\bigwedge_{\mathbb{L}}^2 V$. With an orthogonal pseudobasis, this lattice is an explicit orthogonal sum of three weighted ideal lattices. The decoder is more general: from any matched map $T = \bigwedge^2 C$ and any public pseudobasis of the source module, the spaces $T(e_i \wedge V)$ recover the four image lines. Four rank-one Lenstra–Silverberg computations then recover the restrictions of C , and exact module, Hermitian, determinant, and compound-matrix tests verify their assembly. For the standard cyclotomic orbit, the Hodge lattice is hypercubic and its public S -action reduces shortest-basis matching to a polynomial signed-cycle enumeration.
3. *Half-rank oracle realization.* Only after the preceding direct reductions are complete do we invoke Bambury–Nguyen. In rank two, a calibrated primal–dual lattice has minima product $1/2$ and places the two required involution types in its primal and dual shortest shells; all oracle calls then have rank at most $\lfloor 3d/2 \rfloor + 1$. In rank four, their hypercubic algorithm recovers the required shortest basis using oracle calls of rank at most $3d + 1$. These results reduce oracle rank but do not replace the rank-specific matching and lifting arguments.

The complete SVP-to-Module-LIP reductions are therefore proved for the standard free power-of-two cyclotomic orbit. The fixed-space constructions and the matched-data decoders hold for arbitrary projective modules, and the shape of the rank-four lattice is explicit for every orthogonal pseudobasis. What remains open in the latter setting is to complete generic SVP output on three weighted ideal summands to a list containing the hidden S -linear exterior-square map.

1.2 Technical overview

A common conjugation principle. Both descents start from a canonical conjugate-linear symmetry of the standard Hermitian space. If B is a standard-to-target transition, the public Hermitian form conjugates this standard symmetry by B in rank two and by $\bigwedge^2 B$ in rank four. The fixed space is therefore transported by the hidden transition but can be computed directly from the public Gram matrix. Restricting the ambient tensor metric to this fixed space and intersecting with the appropriate integral module produces the auxiliary Euclidean lattice.

Rank two. The standard symmetry is $\tau_2(v) = J_2 \bar{v}$, where $J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Its diagonal tensor square acts on $\text{Sym}_{\mathbb{L}}^2(V)$. The elementary map $v \otimes w \mapsto vw^t$, after symmetrization and right multiplication by J_2 , sends this action to conjugation on trace-zero endomorphisms. The fixed points are precisely the self-adjoint ones. Cayley–Hamilton then shows that every integral fixed point satisfies $X^2 = qI_2$ with a totally positive algebraic integer q ; the trace metric is minimized exactly when $q = 1$. Thus SVP produces an involution, whose two eigenspaces reduce the original problem to two rank-one isomorphisms.

Rank four. The standard linear Hodge star on $\bigwedge^2 V$ is represented in the standard exterior basis by the matrix Ω pairing complementary indices. Composing it with CM conjugation gives the standard Hermitian involution. For a general determinant-one form, the public Hodge involution is its conjugate by $\bigwedge^2 B$. A matched exterior-square map cannot be recovered from one shortest vector, so the standard cyclotomic argument first obtains a complete shortest basis. The public action of $s = \zeta + \zeta^{-1}$ decomposes this basis into three signed cycles, leaving only a polynomial list of possible S -linear matchings. For each candidate, the four spaces $T(e_i \wedge V)$ determine the image lines of the original basis vectors, after which Lenstra–Silverberg recovery and exact verification reconstruct C .

The resulting proof chains are

$$\begin{aligned} \text{rank two: } & Q \mapsto W_2(Q) \mapsto \mathcal{S}(P, Q), \\ & \mathcal{S}(P, Q) \xrightarrow{\text{SVP}} X \mapsto \text{two rank-one problems} \mapsto C; \\ \text{rank four: } & Q \mapsto W_4(Q) \mapsto \mathcal{W}(P, Q), \\ & \mathcal{W}(P, Q) \xrightarrow{\text{SVP}} \text{a shortest basis} \mapsto T, \\ & T \mapsto \text{four rank-one problems} \mapsto C. \end{aligned}$$

Every final candidate is checked against the original Module-LIP equations; no correctness claim relies solely on an auxiliary-lattice isometry.

Organization. [Section 2](#) fixes the exact input model and recalls the arithmetic, multilinear, and algorithmic facts used later. [Section 3](#) proves the rank-two construction, decoder, direct cyclotomic SVP reduction, and half-rank realization. [Section 4](#) develops the Hodge construction, its diagonal coordinate shape, the general matched-map decoder, and the standard cyclotomic completion. [Section 5](#) summarizes the precise scope of each result. The appendix gives explicit integer-kernel procedures for computing the two auxiliary lattices from public data.

2 Preliminaries

This section records the data needed to make the two descents public and exact. We first fix the CM-field and module conventions. We then show that the public order action and trace Gram matrix recover the full Hermitian form, state the normalized search problem, and recall the symmetric- and exterior-power

constructions used later. The section ends with the two external algorithmic ingredients: rank-one Lenstra–Silverberg recovery and the Bambury–Nguyen oracle-rank reduction.

2.1 CM setting and notation

Throughout, $\mathbb{L}$ is a CM field: a totally imaginary quadratic extension of its maximal totally real subfield $\mathbb{L}^+$. Its nontrivial $\mathbb{L}^+$ -automorphism is complex conjugation $x \mapsto \bar{x}$. We put

$$[\mathbb{L}^+ : \mathbb{Q}] = d, \quad [\mathbb{L} : \mathbb{Q}] = 2d, \quad R = \mathcal{O}_{\mathbb{L}}, \quad S = \mathcal{O}_{\mathbb{L}^+}.$$

An element $a \in \mathbb{L}^+$ is *totally positive*, written $a \gg 0$, if $\sigma(a) > 0$ for every embedding $\sigma : \mathbb{L}^+ \hookrightarrow \mathbb{R}$. The finite group of roots of unity in $\mathbb{L}$ is denoted by $\mu(\mathbb{L})$.

All vectors are column vectors. For a field $\mathbb{F}$, write $M_{m \times n}(\mathbb{F})$ for the m -by- n matrices over $\mathbb{F}$ and $M_r(\mathbb{F}) = M_{r \times r}(\mathbb{F})$. The symbol I_r denotes the r -by- r identity matrix, or simply I when its size is clear; $\text{diag}(a_1, \dots, a_r)$ is the diagonal matrix with the indicated entries. We use $\text{GL}_r(\mathbb{F})$ and $\text{SL}_r(\mathbb{F})$ for the general and special linear groups, and $\perp$ for an orthogonal direct sum in the inner product under discussion. Once bases are fixed, we may use the same letter for a linear map and its coordinate matrix when no ambiguity results. When both levels occur in one argument, $[u]_{\mathcal{E}}$ denotes the coordinate column of u and $[T]_{\mathcal{E}}$ the matrix of T in the ordered basis $\mathcal{E}$; bold letters are reserved for coordinate columns.

For a vector $x = (x_i)$ and a matrix $A = (a_{ij})$ over $\mathbb{L}$, define

$$\bar{x} = (\bar{x}_i), \quad x^* = \bar{x}^t, \quad \bar{A} = (\bar{a}_{ij}), \quad A^t = (a_{ji}), \quad A^* = \bar{A}^t.$$

Thus t denotes ordinary transpose, $\bar{A}$ denotes entrywise CM conjugation, and $*$ denotes conjugate transpose. In particular, if $Q = Q^*$ is Hermitian, then $Q^t = \bar{Q}$.

A map $T : U \rightarrow W$ between $\mathbb{L}$ -spaces is *conjugate-linear* if

$$T(au + bv) = \bar{a}T(u) + \bar{b}T(v) \quad (a, b \in \mathbb{L}).$$

For a finite extension $\mathbb{E}/\mathbb{F}$, the symbols $\text{Tr}_{\mathbb{E}/\mathbb{F}}$ and $N_{\mathbb{E}/\mathbb{F}}$ denote field trace and norm. For an $\mathbb{F}$ -linear endomorphism X , $\text{tr}_{\mathbb{F}}(X)$ and $\det(X)$ denote its ordinary linear trace and determinant; $\text{End}_{\mathbb{F}}(W)$ denotes the $\mathbb{F}$ -algebra of endomorphisms of an $\mathbb{F}$ -space W , and

$$\text{End}_{\mathbb{F}}^0(W) = \{X \in \text{End}_{\mathbb{F}}(W) : \text{tr}_{\mathbb{F}}(X) = 0\}.$$

For a linear or conjugate-linear self-map T , write $\text{Fix}(T) = \{u : T(u) = u\}$; in the conjugate-linear case over $\mathbb{L}/\mathbb{L}^+$, this fixed space is understood as an $\mathbb{L}^+$ -space. For a subset A of an $\mathbb{F}$ -space, $\text{span}_{\mathbb{F}}(A)$ denotes its $\mathbb{F}$ -linear span. Finally, $\langle x, y \rangle$ and $\|x\|$ denote the specified Euclidean inner product and the norm $\|x\| = \sqrt{\langle x, x \rangle}$. Euclidean duals are written with $^\vee$, never with * .

2.2 Hermitian module lattices

A fractional R -ideal is a nonzero finitely generated R -submodule of $\mathbb{L}$ that spans $\mathbb{L}$ over $\mathbb{Q}$. Since R is Dedekind, every fractional ideal is invertible, with

$$\mathfrak{a}^{-1} = \{x \in \mathbb{L} : x\mathfrak{a} \subseteq R\}.$$

We write $\bar{\mathfrak{a}} = \{\bar{a} : a \in \mathfrak{a}\}$.

An R -module lattice of rank r is a finitely generated torsion-free R -module P (equivalently, a finitely generated projective R -module) whose scalar extension $V = \mathbb{L} \otimes_R P$ has $\mathbb{L}$ -dimension r ; as a $\mathbb{Z}$ -module, P has rank $2dr$. Over the Dedekind domain R , such a module has a pseudobasis

$$P = \mathfrak{a}_1 e_1 \oplus \cdots \oplus \mathfrak{a}_r e_r$$

with fractional R -ideals $\mathfrak{a}_i$ and an $\mathbb{L}$ -basis $e_1, \dots, e_r$ of V [4]. In these coordinates,

$$\text{End}_R(P) = \{X = (X_{ij}) \in M_r(\mathbb{L}) : X_{ij}\mathfrak{a}_j \subseteq \mathfrak{a}_i \text{ for all } i, j\}. \quad (1)$$

The pseudobasis is *orthogonal* for a Hermitian form if the e_i are pairwise orthogonal. This is an additional hypothesis in the explicit coordinate families below, not a property of every module lattice.

A Hermitian matrix $Q = Q^*$ defines the form and the Q -adjoint

$$h_Q(x, y) = x^* Q y, \quad X^{\dagger Q} = Q^{-1} X^* Q. \quad (2)$$

Thus $h_Q(Xx, y) = h_Q(x, X^{\dagger Q} y)$; in particular, $X^{\dagger Q} = X$ means self-adjointness for h_Q . The matrix Q , and hence h_Q , is positive definite if, after every embedding $\mathbb{L} \hookrightarrow \mathbb{C}$, the corresponding complex Hermitian matrix is positive definite; we write $Q > 0$. Our convention is that h_Q is conjugate-linear in its first argument and linear in its second. Its determinant then belongs to $\mathbb{L}^+$ and is totally positive. We call a pair (P, Q) with $Q > 0$ a *Hermitian R -module lattice*. Its normalized trace form is

$$g_Q(x, y) = \frac{1}{[\mathbb{L} : \mathbb{Q}]} \text{Tr}_{\mathbb{L}/\mathbb{Q}} h_Q(x, y).$$

On $\mathbb{L}$ -linear maps, the Q -adjoint is also the adjoint for the normalized trace form:

$$g_Q(Xx, y) = g_Q(x, X^{\dagger Q} y). \quad (3)$$

Thus, if G and $X_{\mathbb{Q}}$ are the rational matrices of g_Q and X in a $\mathbb{Z}$ -basis of P , then

$$X^{\dagger Q} = X \iff X_{\mathbb{Q}}^t G = G X_{\mathbb{Q}}.$$

Here the superscript t is the ordinary transpose over $\mathbb{Q}$, as in the matrix convention above.

For a fractional ideal $\mathfrak{a}$ and $a \in \mathbb{L}^+$ with $a \gg 0$, the notation $(\mathfrak{a}, a)$ denotes the rank-one Hermitian R -lattice with underlying module $\mathfrak{a}$ and form $h(x, y) = a\bar{x}y$.

Exact public representation. We use two equivalent exact presentations. In the algebraic presentation, R is specified by a $\mathbb{Z}$ -basis, its multiplication table, and the matrix of conjugation; P is specified by a pseudobasis, and h_Q by its Hermitian matrix in fixed $\mathbb{L}$ -coordinates. In the Euclidean presentation, one gives a $\mathbb{Z}$ -basis of P , the action matrices of a $\mathbb{Z}$ -basis of R , and the rational Gram matrix of g_Q . These are the standard number-field and module encodings of [4]; the latter presentation is the public interface used in Module-LIP algorithms [5,12]. Neither presentation includes a hidden isometry. The next theorem shows directly that the Euclidean presentation recovers the algebraic one.

Let

$$\rho : R \longrightarrow \text{End}_{\mathbb{Z}}(P)$$

denote the public action. Its compatibility with the trace metric is

$$g_Q(\rho(a)x, y) = g_Q(x, \rho(\bar{a})y).$$

Theorem 1 (Trace-dual reconstruction). *Let $\omega_1, \dots, \omega_{2d}$ be a $\mathbb{Z}$ -basis of R , hence a $\mathbb{Q}$ -basis of $\mathbb{L}$, and let $\eta_1, \dots, \eta_{2d}$ be its conjugate trace-dual basis:*

$$\text{Tr}_{\mathbb{L}/\mathbb{Q}}(\bar{\omega}_i \eta_j) = \delta_{ij}.$$

Then

$$h_Q(x, y) = [\mathbb{L} : \mathbb{Q}] \sum_{j=1}^{2d} g_Q(\omega_j x, y) \eta_j \quad (x, y \in P).$$

Thus the public module action and trace Gram matrix determine h_Q in deterministic polynomial time. Evaluating the reconstructed form on a public $\mathbb{L}$ -basis of V recovers Q exactly; when the basis vectors do not lie in P , one first clears a common denominator.

Proof. If $h_Q(x, y) = \sum_j c_j \eta_j$, then

$$[\mathbb{L} : \mathbb{Q}] g_Q(\omega_i x, y) = \text{Tr}_{\mathbb{L}/\mathbb{Q}}(\bar{\omega}_i h_Q(x, y)) = c_i.$$

The remaining operations are rational linear algebra in fixed number-field bases.

Consequently, a rational matrix $C_{\mathbb{Q}}$ between the public $\mathbb{Z}$ -bases is an R -module isometry precisely when

$$C_{\mathbb{Q}} \in \text{GL}_{2dr}(\mathbb{Z}), \quad C_{\mathbb{Q}} \rho_0(a) = \rho_1(a) C_{\mathbb{Q}}, \quad C_{\mathbb{Q}}^t G_1 C_{\mathbb{Q}} = G_0,$$

where it suffices to check the middle equation on a $\mathbb{Z}$ -basis of R . Thus every Hermitian and module equation used below is an exact public test. We now state the normalized search problem to which the two descents apply.

2.3 Normalized Module-LIP and determinant normalization

Fix public $\mathbb{L}$ -coordinates $V = \mathbb{L}^r$. The matrices Q_i and their determinants below always refer to these coordinates.

Definition 2 (Normalized Module-LIP). *The determinant-one, special-linear Module-LIP instance consists of two pairs (P_0, Q_0) and (P_1, Q_1) satisfying*

$$\det Q_0 = \det Q_1 = 1.$$

It is promised that there exists an $\mathbb{L}$ -linear map C such that

$$C(P_0) = P_1, \quad C^* Q_1 C = Q_0, \quad \det C = 1,$$

and the task is to recover one such C .

Here (P_i, Q_i) denotes the Hermitian modules recovered from either exact presentation above. Determinant one is a condition in the fixed public $\mathbb{L}$ -coordinates. The normalization is a restriction on the represented search problem, not an assertion that an arbitrary determinant has an r th root in $\mathbb{L}^+$. If a free-module instance is written as $Q_1 = U^* Q_0 U$, then the corresponding map in [definition 2](#) is $C = U^{-1}$.

Theorem 3 (Lenstra–Silverberg [11, Theorem 1.3]). *Let $\mathfrak{J}$ be a fractional R -ideal and let $w \in \mathbb{L}^+$ be totally positive with*

$$\mathfrak{J}\bar{\mathfrak{J}} = Rw.$$

There is a deterministic polynomial-time algorithm that decides whether there is $v \in \mathbb{L}^\times$ satisfying

$$\mathfrak{J} = Rv, \quad v\bar{v} = w,$$

and returns such a v when it exists. If v is one solution, then all solutions are $v\mu(\mathbb{L})$ [11, Theorem 1.3 and Proposition 7.4].

For rank-one Hermitian lattices $(\mathfrak{J}_0, q_0)$ and $(\mathfrak{J}_1, q_1)$, set

$$\mathfrak{J} = \mathfrak{J}_1 \mathfrak{J}_0^{-1}, \quad w = \frac{q_0}{q_1}.$$

A promised isometry is recovered by checking $\mathfrak{J}\bar{\mathfrak{J}} = Rw$ and applying the preceding algorithm. The quotient w is totally positive because $q_0, q_1 \gg 0$. Its output α satisfies $\mathfrak{J}_1 = \alpha \mathfrak{J}_0$ and $q_1 \alpha \bar{\alpha} = q_0$, so multiplication by α is the required isometry.

Proposition 4 (Free-module special-linear normalization). *Let $P = R^r$ and suppose*

$$Q_1 = U^* Q_0 U \quad \text{for some } U \in \text{GL}_r(R).$$

Put $\alpha = \det U$. From the public value

$$w = \frac{\det Q_1}{\det Q_0} = \bar{\alpha} \alpha \quad \text{and} \quad \alpha R = R,$$

the Lenstra–Silverberg norm-constrained generator algorithm, applied with $\mathfrak{J} = R$, returns one unit β_0 satisfying $\beta_0 \bar{\beta}_0 = w$. All solutions form the finite coset $\beta_0 \mu(\mathbb{L}) = \alpha \mu(\mathbb{L})$. For each candidate β in that coset, let $D_\beta = \text{diag}(\beta^{-1}, 1, \dots, 1)$ and $Q_{1,\beta} = D_\beta^* Q_1 D_\beta$. Then $\beta \in R^\times$, $D_\beta \in \text{GL}_r(R)$, and

$$\det Q_{1,\beta} = \det Q_0, \quad Q_{1,\beta} = (UD_\beta)^* Q_0 (UD_\beta).$$

One branch has $\beta = \alpha$ and hence $\det(UD_\beta) = 1$. Consequently, when $\det Q_0 = 1$, solving and verifying the finitely many normalized instances recovers a solution of the original free-module instance. More explicitly, if a normalized branch returns C_β with $C_\beta^* Q_{1,\beta} C_\beta = Q_0$, then $D_\beta C_\beta$ is the corresponding candidate for the original instance.

Proof. Apply [theorem 3](#) with $\mathfrak{J} = R$. Every root of unity has relative norm one, so all candidates have $\bar{\beta}\beta = \bar{\alpha}\alpha$. The displayed determinant and congruence identities follow directly. The group $\mu(\mathbb{L})$ can be computed in polynomial time [[13](#), Lemma 2.5]; enumerating it and checking the original module and Hermitian equations retains exactly the valid branches. This is the free-module determinant normalization used for Module-LIP in [[12](#), Section 2.5].

2.4 Symmetric and exterior powers

The hidden transition will be represented on a smaller tensor space: the symmetric square in rank two and the second exterior power in rank four. We fix here the common multilinear conventions and integral structures; the rank-specific involutions and metrics are introduced in [sections 3](#) and [4](#).

Let $\iota(v \otimes w) = w \otimes v$ be the swap map on $V \otimes_{\mathbb{L}} V$. Put

$$\text{Sym}_{\mathbb{L}}^2(V) = (V \otimes_{\mathbb{L}} V) / \text{span}_{\mathbb{L}} \{v \otimes w - w \otimes v : v, w \in V\},$$

and write $v \odot w$ for the class of $(v \otimes w + w \otimes v)/2$; in particular, $v^2 = v \odot v$. Since the characteristic is zero, $\text{Sym}_{\mathbb{L}}^2(V)$ may equivalently be identified with the $+1$ eigenspace of ι , and

$$\dim_{\mathbb{L}} \text{Sym}_{\mathbb{L}}^2(V) = \frac{r(r+1)}{2}.$$

Lemma 5 (Diagonal tensor action). *For any linear or conjugate-linear map $A : V \rightarrow V$, the diagonal action $A \otimes A$ commutes with ι . Hence the subspace generated by $v \otimes w - w \otimes v$ is stable, and $A \otimes A$ induces a map on $\text{Sym}_{\mathbb{L}}^2(V)$ with the same linearity type as A . If $A^2 = -I$, then the induced map is an involution.*

Proof. If A is conjugate-linear, the rule is well defined over $\mathbb{L}$ because

$$A(av) \otimes A(w) = \bar{a} A(v) \otimes A(w) = A(v) \otimes \bar{a} A(w) = A(v) \otimes A(aw).$$

It is then conjugate-linear on $V \otimes_{\mathbb{L}} V$; the linear case is immediate. Both composites with ι send $v \otimes w$ to $Aw \otimes Av$. Moreover, $(A \otimes A)^2 = A^2 \otimes A^2 = I$.

Exterior powers and integral structures. For $1 \leq k \leq r$, the exterior power $\bigwedge_{\mathbb{L}}^k V$ is the alternating quotient of $V^{\otimes_{\mathbb{L}} k}$, and $\dim_{\mathbb{L}} \bigwedge_{\mathbb{L}}^k V = \binom{r}{k}$. In particular,

$$\bigwedge_{\mathbb{L}}^2 V = (V \otimes_{\mathbb{L}} V) / \text{span}_{\mathbb{L}} \{v \otimes v : v \in V\},$$

and $v \wedge w$ denotes the class of $v \otimes w$. For P , define

$$\bigwedge_R^k P = \text{span}_R \{x_1 \wedge \cdots \wedge x_k : x_i \in P\}.$$

It has rank $\binom{r}{k}$ and, in the pseudobasis above,

$$\bigwedge_R^k P = \bigoplus_{i_1 < \cdots < i_k} \mathbf{a}_{i_1} \cdots \mathbf{a}_{i_k} e_{i_1} \wedge \cdots \wedge e_{i_k}. \quad (4)$$

In particular, the determinant line $\det_R(P) = \bigwedge_R^r P$ is an invertible rank-one R -module. Its R -endomorphism ring is R , and the map induced there by $X \in \text{End}_R(P)$ is multiplication by $\det X$. This is the integrality principle used later for determinants of public endomorphisms.

Every linear or conjugate-linear map $A : V \rightarrow V$ induces diagonal maps $\text{Sym}^2(A)$ and $\bigwedge^k A$; if $A(P_0) = P_1$, then

$$\bigwedge^k A \left(\bigwedge_R^k P_0 \right) = \bigwedge_R^k P_1.$$

For a fixed basis $e_1, \dots, e_r$, we abbreviate $e_{ij} = e_i \wedge e_j$ for $i < j$.

The fixed spaces in both descents recover their ambient $\mathbb{L}$ -spaces after extending scalars from $\mathbb{L}^+$ to $\mathbb{L}$. We use the following elementary fact repeatedly.

Lemma 6 (Fixed space of a conjugate-linear involution). *Let U be an n -dimensional $\mathbb{L}$ -space and let $\tau : U \rightarrow U$ be a conjugate-linear involution. Then*

$$W = \text{Fix}(\tau)$$

has dimension n over $\mathbb{L}^+$, and scalar extension gives

$$\mathbb{L} \otimes_{\mathbb{L}^+} W \simeq U.$$

Proof. Choose $\vartheta \in \mathbb{L}^\times$ with $\bar{\vartheta} = -\vartheta$. Every $u \in U$ has the unique decomposition

$$u = \frac{u + \tau(u)}{2} + \vartheta \frac{u - \tau(u)}{2\vartheta} \in W \oplus \vartheta W.$$

For rank four, we also need the standard matrix realization of $\bigwedge_{\mathbb{L}}^2 V$. For $M \in M_4(\mathbb{L})$, its second compound matrix $C_2(M)$ is the matrix of $\bigwedge^2 M$ in the ordered exterior basis

$$\mathcal{E}_2 = (e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34});$$

equivalently, its (ij, kl) entry is the minor $\det M_{\{i,j\},\{k,l\}}$. Compound matrices satisfy

$$\begin{aligned} C_2(AB) &= C_2(A)C_2(B), & C_2(M^t) &= C_2(M)^t, \\ C_2(\bar{M}) &= \overline{C_2(M)}, & C_2(M^*) &= C_2(M)^*. \end{aligned}$$

The Hermitian form induced by Q on $\bigwedge_{\mathbb{L}}^2 V$ is

$$h_{Q,2}(x_1 \wedge x_2, y_1 \wedge y_2) = \det(h_Q(x_i, y_j))_{1 \leq i,j \leq 2},$$

extended sesquilinearly. Its matrix is $Q_2 = C_2(Q)$; in particular, $Q_2 = Q_2^*$ and $Q_2 > 0$. Thus $\bigwedge^2 C$ transports both the induced Hermitian form and the integral exterior-power module whenever C transports (P_0, Q_0) to (P_1, Q_1) .

2.5 Euclidean lattices and prior reductions

A Euclidean lattice is a discrete free $\mathbb{Z}$ -submodule Λ of finite rank m in a finite-dimensional real inner-product space. Its dual, always taken in $\text{span}_{\mathbb{R}}(\Lambda)$ with the specified inner product, is

$$\Lambda^\vee = \{x \in \text{span}_{\mathbb{R}}(\Lambda) : \langle x, y \rangle \in \mathbb{Z} \text{ for every } y \in \Lambda\}.$$

If G is the Gram matrix in a basis of Λ , the dual basis has Gram matrix G^{-1} . The first minimum and shortest shell are

$$\lambda_1(\Lambda) = \min \{\|x\| : 0 \neq x \in \Lambda\}, \quad \text{Min}(\Lambda) = \{x \in \Lambda : \|x\| = \lambda_1(\Lambda)\}.$$

A γ -SVP oracle returns a nonzero $x \in \Lambda$ with $\|x\| \leq \gamma \lambda_1(\Lambda)$; exact SVP is the case $\gamma = 1$. We call Λ *hypercubic* if it has an orthogonal $\mathbb{Z}$ -basis whose vectors all have the same norm. This permits the common scaling of the orthonormal-basis convention in [2]; normalizing that common length does not change an approximation factor.

All auxiliary lattices in this paper are represented by exact rational Gram matrices. We first clear a common positive denominator, so that the Gram matrix is integral. This is the basis-Gram form of the input used below: the proof of [2, Theorem 1] uses exact Gram operations and the integrality of squared sublattice covolumes, both of which hold for an integral Gram matrix. Replacing G by cG for $c > 0$ preserves the ordering of primal lengths, multiplies $\lambda_1(\Lambda)$ by $\sqrt{c}$, and multiplies $\lambda_1(\Lambda^\vee)$ by $c^{-1/2}$. Hence approximation factors and the primal-dual minima product are unchanged.

Power-of-two cyclotomic coordinates. The two explicit free-module families use the following common coefficient model. Let

$$\mathbb{L} = \mathbb{Q}[X]/(X^N + 1), \quad R = \mathbb{Z}[X]/(X^N + 1),$$

where $N = [\mathbb{L} : \mathbb{Q}] = 2d$ is a power of two, and write ζ for the image of X ; thus $\bar{\zeta} = \zeta^{-1}$. For $f = \sum_{i=0}^{N-1} a_i \zeta^i$, put

$$\text{vec}(f) = (a_0, \dots, a_{N-1})^t.$$

Then

$$\langle f, g \rangle_{\text{coeff}} = \text{vec}(f)^t \text{vec}(g) = \frac{1}{N} \text{Tr}_{\mathbb{L}/\mathbb{Q}}(\bar{f}g), \quad (5)$$

so $1, \zeta, \dots, \zeta^{N-1}$ is an orthonormal $\mathbb{Z}$ -basis of R . Put

$$s = \zeta + \zeta^{-1}, \quad s_i = \zeta^i + \zeta^{-i} \quad (1 \leq i < d).$$

Since $\zeta^{-i} = -\zeta^{N-i}$, taking the conjugation-fixed part of the coefficient basis gives

$$S = \mathbb{Z} \cdot 1 \oplus \bigoplus_{i=1}^{d-1} \mathbb{Z} \cdot s_i.$$

The recurrence $ss_i = s_{i+1} + s_{i-1}$, with $s_0 = 2$, shows inductively that $S = \mathbb{Z}[s]$. Moreover, [equation \(5\)](#) shows that $1, s_1, \dots, s_{d-1}$ is an orthogonal $\mathbb{Z}$ -basis of S with squared lengths $1, 2, \dots, 2$ for the normalized trace form on $\mathbb{L}^+$.

Theorem 7 (Bambury–Nguyen primal–dual reduction [2, Theorem 1, inequality (3), and Theorem 4]). *Let Λ be a rank- m Euclidean lattice, and let $b \geq \lambda_1(\Lambda)$ and $b^\vee \geq \lambda_1(\Lambda^\vee)$. If*

$$\lambda_1(\Lambda)\lambda_1(\Lambda^\vee) < 1 - \varepsilon, \quad \varepsilon \geq \frac{1}{\text{poly}(m)},$$

then the primal–dual block algorithm of Bambury and Nguyen returns a nonzero $x \in \Lambda$ with $\|x\| \leq b$ or a nonzero $y \in \Lambda^\vee$ with $\|y\| \leq b^\vee$. Its number of iterations is polynomial in the input bit length and $1/\varepsilon$. Each iteration makes four exact-SVP calls in ranks

$$\lfloor m/2 \rfloor, \quad m - \lfloor m/2 \rfloor, \quad m - \lfloor m/2 \rfloor - 1, \quad \lfloor m/2 \rfloor + 1.$$

In particular, every call has rank at most

$$\boxed{\lfloor m/2 \rfloor + 1}.$$

Their inequality (3) permits the two reduction calls to use approximation factors $\gamma, \gamma^\vee$ whenever

$$\gamma\gamma^\vee\lambda_1(\Lambda)\lambda_1(\Lambda^\vee) < 1 - \varepsilon;$$

the two shell-detection calls remain exact. For a rank- m hypercubic lattice, their specialized algorithm allows all calls to use a γ -SVP oracle and recovers an orthogonal equal-length basis in rank at most $\lfloor m/2 \rfloor + 1$ for $\gamma < \sqrt{2 - 2/m}$.

3 Rank two: symplectic descent through $\text{Sym}_{\mathbb{L}}^2(V)$

We now specialize the normalized problem of [definition 2](#) to module rank two. The public endpoints are Hermitian R -modules (P_i, Q_i) with

$$V_i = \mathbb{L} \otimes_R P_i \simeq \mathbb{L}^2, \quad \text{rank}_{\mathbb{Z}} P_i = 4d, \quad \det Q_i = 1.$$

The task is to recover an $\mathbb{L}$ -linear map

$$C(P_0) = P_1, \quad C^*Q_1C = Q_0, \quad \det C = 1.$$

All matrices and determinants refer to the displayed public coordinates. We write (P, Q) when discussing one endpoint.

The argument proceeds in four steps. We first conjugate the standard symplectic symmetry by the public Hermitian form and pass to a three-dimensional fixed space. We then identify this space with trace-zero self-adjoint endomorphisms, choose its integral lattice and invariant metric, and show that its shortest vectors are involutions. Next we prove an exact decoder from a matched involution pair to the original transition. Only in the final two subsections do we specialize to the standard cyclotomic reference

$$(R^2, I_2) \longrightarrow (R^2, Q), \quad B \in \mathrm{SL}_2(R), \quad B^*QB = I_2,$$

where the source shell is explicit and the SVP calls can be reduced to essentially half rank. Here B denotes a standard-to-target transition, whereas C denotes a transition between arbitrary endpoints.

3.1 The canonical symplectic symmetry and its public conjugate

Put

$$J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \tau_2(v) = J_2 \bar{v}.$$

Thus τ_2 is conjugate-linear and $\tau_2^2 = -I$. The elementary rank-two symplectic identity is

$$M^t J_2 M = (\det M) J_2 \quad (M \in M_2(\mathbb{L})). \quad (6)$$

For a determinant-one Hermitian matrix Q , define the public conjugate-linear map

$$J_Q(v) = J_2 \bar{Q} \bar{v}. \quad (7)$$

Proposition 8 (Public conjugate of the standard symmetry). *The map J_Q satisfies*

$$J_Q^2 = -I, \quad h_Q(J_Q v, J_Q w) = \overline{h_Q(v, w)}.$$

If $B \in \mathrm{SL}_2(\mathbb{L})$ is a normalized transition from the standard form to Q , namely

$$B^*QB = I_2,$$

then

$$\boxed{J_Q = B \circ \tau_2 \circ B^{-1}.} \quad (8)$$

*More generally, if $C^*Q_1C = Q_0$ and $\det C = 1$, then*

$$C J_{Q_0} = J_{Q_1} C. \quad (9)$$

Proof. Since $Q = Q^*$, one has $Q^t = \bar{Q}$. Applying [equation \(6\)](#) to Q gives $\bar{Q}J_2Q = J_2$, and hence

$$J_Q^2 = J_2\bar{Q}J_2Q = J_2^2 = -I.$$

The Hermitian identity follows by substitution. If $B^*QB = I_2$, then $Q = (B^{-1})^*B^{-1}$ and $\bar{Q} = B^{-t}\bar{B}^{-1}$. Applying [equation \(6\)](#) to B^t gives $BJ_2B^t = J_2$, or equivalently $J_2B^{-t} = BJ_2$. Therefore

$$J_Q(v) = J_2B^{-t}\bar{B}^{-1}\bar{v} = BJ_2\overline{B^{-1}v} = B\tau_2(B^{-1}v).$$

Finally, from $Q_0 = C^*Q_1C$ we obtain $\bar{Q}_0 = C^t\bar{Q}_1\bar{C}$; using $CJ_2C^t = J_2$ gives [equation \(9\)](#).

Equivalently, if $D = B^{-1}$ is the Gram factor, so that $Q = D^*D$, then

$$J_Q = D^{-1} \circ \tau_2 \circ D.$$

This is the same operator written in Gram-factor rather than standard-to-target coordinates.

The diagonal tensor action of J_Q commutes with the swap of the two tensor factors. We therefore obtain a conjugate-linear involution on the symmetric square:

$$\Sigma_Q = -(J_Q \otimes J_Q)|_{\text{Sym}_{\mathbb{L}}^2(V)}. \quad (10)$$

The minus sign is forced by the self-adjoint matrix model below.

Definition 9 (Rank-two fixed space).

$$W_2(Q) = \text{Fix}(\Sigma_Q) \subseteq \text{Sym}_{\mathbb{L}}^2(V).$$

Proposition 10 (Conjugated fixed space). *The space $W_2(Q)$ has dimension three over $\mathbb{L}^+$, and*

$$\mathbb{L} \otimes_{\mathbb{L}^+} W_2(Q) \simeq \text{Sym}_{\mathbb{L}}^2(V).$$

*If $B \in \text{SL}_2(\mathbb{L})$ satisfies $B^*QB = I_2$, then*

$$\Sigma_Q = \text{Sym}^2(B)\Sigma_{I_2}\text{Sym}^2(B)^{-1}, \quad W_2(Q) = \text{Sym}^2(B)W_2(I_2).$$

Proof. The first assertion is [lemma 6](#). The conjugation formula follows from [equation \(8\)](#) by taking the diagonal tensor square and passing to the symmetric quotient.

3.2 From symmetric tensors to trace-zero endomorphisms

The fixed space above is natural but still abstract. To read its short vectors as algebraic information about the original module, we now identify it with a familiar endomorphism space. The coordinate map used below is obtained from the standard tensor identification, not chosen ad hoc. In the fixed public basis let

$$\Theta : V \otimes_{\mathbb{L}} V \longrightarrow M_2(\mathbb{L}), \quad \Theta(v \otimes w) = vw^t.$$

The map Θ itself does not descend through the symmetric quotient. Instead, using the canonical symmetrization section in characteristic zero, we obtain the $\mathbb{L}$ -linear isomorphism

$$\kappa : \text{Sym}_{\mathbb{L}}^2(V) \longrightarrow \{S \in M_2(\mathbb{L}) : S^t = S\}, \quad \kappa(v \odot w) = \frac{vw^t + wv^t}{2}.$$

Thus, for $u = xe_1^2 + 2y(e_1 \odot e_2) + ze_2^2$,

$$\kappa(u) = \begin{pmatrix} x & y \\ y & z \end{pmatrix}.$$

Define

$$\boxed{\Phi(u) = \kappa(u)J_2.} \tag{11}$$

In coordinates this is precisely

$$\Phi(u) = \begin{pmatrix} -y & x \\ -z & y \end{pmatrix}. \tag{12}$$

Right multiplication by J_2 is the essential step: $\kappa(u)$ transforms by congruence, while $\Phi(u)$ transforms by conjugation.

Theorem 11 (Symmetric-square matrix model). *The map*

$$\Phi : \text{Sym}_{\mathbb{L}}^2(V) \longrightarrow \text{End}_{\mathbb{L}}^0(V)$$

is an $\mathbb{L}$ -linear isomorphism. For every $u \in \text{Sym}_{\mathbb{L}}^2(V)$,

$$\boxed{\Phi(\Sigma_Q u) = \Phi(u)^{\dagger_Q}.} \tag{13}$$

Consequently

$$\Phi(W_2(Q)) = \{X \in \text{End}_{\mathbb{L}}(V) : X^{\dagger_Q} = X, \text{tr}_{\mathbb{L}} X = 0\}.$$

For $C \in \text{SL}_2(\mathbb{L})$,

$$\boxed{\Phi(\text{Sym}^2(C)u) = C\Phi(u)C^{-1}.} \tag{14}$$

*In particular, if $C^*Q_1C = Q_0$, then $\text{Sym}^2(C)W_2(Q_0) = W_2(Q_1)$.*

Proof. A symmetric matrix S is sent to SJ_2 , and $\text{tr}(SJ_2) = 0$. Conversely, if $X \in M_2(\mathbb{L})$ has trace zero, then $-XJ_2$ is symmetric, so Φ is an isomorphism.

For $C \in \text{SL}_2(\mathbb{L})$,

$$\kappa(\text{Sym}^2(C)u) = C\kappa(u)C^t \quad \text{and} \quad C^tJ_2 = J_2C^{-1}.$$

Therefore

$$\Phi(\text{Sym}^2(C)u) = C\kappa(u)C^tJ_2 = C\Phi(u)C^{-1},$$

which proves [equation \(14\)](#).

Let $S = \kappa(u)$. Since J_Q has semilinear matrix $J_2 \bar{Q}$, the induced action on symmetric matrices is

$$\kappa((J_Q \otimes J_Q)u) = J_2 \bar{Q} \bar{S} (J_2 \bar{Q})^t.$$

Using $(J_2 \bar{Q})^t = -Q J_2$, the definition $\Sigma_Q = -(J_Q \otimes J_Q)$ gives

$$\Phi(\Sigma_Q u) = -J_2 \bar{Q} \bar{S} Q.$$

On the other hand, applying [equation \(6\)](#) to $\bar{Q}$ yields $Q J_2 \bar{Q} = J_2$, hence $Q^{-1} J_2 = J_2 \bar{Q}$. Therefore

$$\Phi(u)^{\dagger_Q} = Q^{-1} (S J_2)^* Q = -Q^{-1} J_2 \bar{S} Q = -J_2 \bar{Q} \bar{S} Q,$$

which proves [equation \(13\)](#) directly. The fixed-space description follows immediately.

3.3 Metric normalization and the integral lattice

The preceding identification determines the algebraic fixed space. An SVP reduction additionally needs a full integral lattice and a Euclidean metric that are both public and preserved by the hidden transition. We obtain them from the ambient symmetric-square metric rather than from an arbitrary coordinate norm.

Let $h_{Q,\text{sym}}$ be the Hermitian form induced by h_Q on $\text{Sym}_{\mathbb{L}}^2(V)$; on pure symmetric tensors,

$$h_{Q,\text{sym}}(v_1 \odot v_2, w_1 \odot w_2) = \frac{1}{2} (h_Q(v_1, w_1) h_Q(v_2, w_2) + h_Q(v_1, w_2) h_Q(v_2, w_1)).$$

Proposition 12 (Tensor metric versus trace metric). *For $u, v \in W_2(Q)$,*

$$h_{Q,\text{sym}}(u, v) = \text{tr}_{\mathbb{L}}(\Phi(u)\Phi(v)) \in \mathbb{L}^+. \quad (15)$$

We equip $W_2(Q)$ with

$$\langle u, v \rangle_2 = \frac{1}{2[\mathbb{L} : \mathbb{Q}]} \text{Tr}_{\mathbb{L}/\mathbb{Q}} h_{Q,\text{sym}}(u, v) = \frac{1}{4d} \text{Tr}_{\mathbb{L}/\mathbb{Q}} \text{tr}_{\mathbb{L}}(\Phi(u)\Phi(v)). \quad (16)$$

Thus our Euclidean metric is one half of the normalized trace of the ambient tensor metric.

Proof. After any embedding $\mathbb{L} \hookrightarrow \mathbb{C}$, choose the positive square root $D = Q^{1/2}$; since $\det Q = 1$, it satisfies $D \in \text{SL}_2(\mathbb{C})$ and $Q = D^* D$. The map $\text{Sym}^2(D)$ sends $W_2(Q)$ to $W_2(I_2)$ and is an isometry from the Q -tensor metric to the standard tensor metric, while [equation \(14\)](#) transports Φ by conjugation. It is therefore enough to check $Q = I_2$. There an element of the fixed space corresponds to

$$X = \begin{pmatrix} a & z \\ \bar{z} & -a \end{pmatrix}, \quad a \in \mathbb{L}^+,$$

and its symmetric tensor has coefficient matrix $-X J_2$. Direct expansion of the tensor metric gives $h_{I_2,\text{sym}}(u, v) = \text{tr}_{\mathbb{L}}(XY)$. Equality under every embedding gives [equation \(15\)](#) over $\mathbb{L}$.

Remark 13 (Why the factor $1/2$?). The tensor-induced metric is canonical and isometry-invariant. The factor $1/2$ in [equation \(16\)](#) is only a global normalization, so it does not change shortest vectors, approximation factors, or the product of primal and dual minima. It makes the fixed-coordinate model below carry the ordinary coefficient norm: the two conjugate tensor coordinates are counted once rather than twice. A coordinate Euclidean metric chosen before the fixed-space restriction would not be invariant under the hidden transition and is therefore not used in the reduction.

Definition 14 (Rank-two auxiliary lattice).

$$\boxed{\mathcal{S}(P, Q) = \{u \in W_2(Q) : \Phi(u)P \subseteq P\}}. \quad (17)$$

It is equipped with [equation \(16\)](#).

Proposition 15 (Integral transport). *The lattice $\mathcal{S}(P, Q)$ has $\mathbb{Z}$ -rank $3d$. If*

$$C(P_0) = P_1, \quad C^*Q_1C = Q_0, \quad \det C = 1,$$

then

$$\mathrm{Sym}^2(C) : \mathcal{S}(P_0, Q_0) \xrightarrow{\sim} \mathcal{S}(P_1, Q_1)$$

is a Euclidean isometry.

Proof. The fixed space has $\mathbb{L}^+$ -dimension three. In the exact public representation, self-adjointness and trace zero are rational linear equations on $\mathrm{End}_R(P) \otimes_{\mathbb{Z}} \mathbb{Q}$; hence their integral solutions form a full $\mathbb{Z}$ -lattice of rank $3d$. Integrality is preserved by [equation \(14\)](#); the metric is preserved by the tensor isometry, or equivalently by invariance of matrix trace under conjugation.

For $u \in W_2(Q)$, write $X = \Phi(u)$. Cayley–Hamilton gives

$$X^2 = q_Q(u)I_2, \quad q_Q(u) = -\det X = \frac{1}{2} \mathrm{tr}_{\mathbb{L}}(X^2) \in \mathbb{L}^+, \quad (18)$$

and

$$\|u\|_2^2 = \frac{1}{d} \mathrm{Tr}_{\mathbb{L}^+/\mathbb{Q}} q_Q(u).$$

Theorem 16 (Unit shell). *For every nonzero $u \in \mathcal{S}(P, Q)$,*

$$\|u\|_2^2 \geq 1.$$

Equality holds if and only if

$$\Phi(u)^2 = I_2.$$

Thus, whenever the lattice contains an integral self-adjoint involution, its shortest vectors are exactly such involutions.

Proof. Because X preserves the rank-two projective module P , its determinant acts integrally on the invertible module $\bigwedge_R^2 P$. Hence $q_Q(u) \in R$. Self-adjointness fixes the determinant under conjugation, so $q_Q(u) \in S$. At every real embedding of $\mathbb{L}^+$, the trace-zero self-adjoint matrix X has eigenvalues $\lambda, -\lambda$; therefore every conjugate of $q_Q(u)$ is positive for $u \neq 0$. The arithmetic–geometric mean inequality gives

$$\frac{1}{d} \operatorname{Tr}_{\mathbb{L}^+/\mathbb{Q}} q_Q(u) \geq N_{\mathbb{L}^+/\mathbb{Q}}(q_Q(u))^{1/d} \geq 1.$$

Equality forces all conjugates to equal one, hence $q_Q(u) = 1$ and [equation \(18\)](#) gives $X^2 = I_2$. Conversely, $X^2 = I_2$ gives $q_Q(u) = 1$ and therefore $\|u\|_2^2 = 1$.

Proposition 17 (Rank-two coordinate shape). *Assume that*

$$P = \mathfrak{a}_1 e_1 \oplus \mathfrak{a}_2 e_2, \quad Q = \operatorname{diag}(\alpha_1, \alpha_2), \quad \alpha_1 \alpha_2 = 1$$

is an orthogonal pseudobasis. Every element of $\Phi(S(P, Q))$ is uniquely

$$A(a, z) = \begin{pmatrix} a & z \\ (\alpha_1/\alpha_2)\bar{z} & -a \end{pmatrix}, \quad (19)$$

where

$$a \in S, \quad z \in \mathfrak{I}_{12} := \mathfrak{a}_1 \mathfrak{a}_2^{-1} \cap \frac{\alpha_2}{\alpha_1} \bar{\mathfrak{a}}_2 \bar{\mathfrak{a}}_1^{-1}. \quad (20)$$

Moreover,

$$A(a, z)^2 = \left(a^2 + \frac{\alpha_1}{\alpha_2} z \bar{z} \right) I_2, \quad q_Q(a, z) = a^2 + \frac{\alpha_1}{\alpha_2} z \bar{z}. \quad (21)$$

Consequently $S(P, Q)$ is the orthogonal direct sum of the S -coordinate a and the fractional-ideal coordinate z , with the metric weights displayed in [equation \(21\)](#).

Proof. Write a trace-zero endomorphism as $X = \begin{pmatrix} a & z \\ w & -a \end{pmatrix}$. The equation $X^{\dagger_Q} = X$ is equivalent to

$$a = \bar{a}, \quad w = \frac{\alpha_1}{\alpha_2} \bar{z}.$$

The diagonal integrality conditions in [equation \(1\)](#) then give $a \in R \cap \mathbb{L}^+ = S$. The upper-right condition gives $z \in \mathfrak{a}_1 \mathfrak{a}_2^{-1}$, while integrality of the lower-left entry gives

$$\frac{\alpha_1}{\alpha_2} \bar{z} \in \mathfrak{a}_2 \mathfrak{a}_1^{-1},$$

which is equivalent to the second ideal condition in [equation \(20\)](#). Multiplying the matrix in [equation \(19\)](#) gives [equation \(21\)](#); polarization shows that the two coordinate summands are orthogonal for [equation \(16\)](#).

3.4 Recovering Module-LIP from matched shortest involutions

The unit-shell theorem explains what an auxiliary SVP solution represents. We now show that a correctly matched source–target pair of such vectors is already sufficient to recover the original Module-LIP transition.

For an integral self-adjoint involution X , the rational decomposition $V = V_+ \oplus V_-$ need not restrict to an integral direct-sum decomposition of P . Indeed, with $P_\pm = P \cap V_\pm$, one always has

$$2P \subseteq P_+ \oplus P_- \subseteq P,$$

because $(I + X)p$ and $(I - X)p$ lie in the two eigenline modules and sum to $2p$. We do not need to represent the resulting finite quotient separately: the decoder below assembles all rank-one branches and tests the original equality $C(P_0) = P_1$ directly.

The use of a non-scalar automorphism to expose rank-one submodules follows the eigenspace strategy of [13,12]. We state the exact CM-order reduction in the general pseudobasis input model because it is the endpoint of the auxiliary SVP reduction.

Proposition 18 (Decoder for a candidate involution pair). *Let (P_i, Q_i) be a normalized rank-two Module-LIP instance, and let $X_i \in \text{End}_R(P_i)$ satisfy*

$$X_i^{\dagger Q_i} = X_i, \quad \text{tr}_{\mathbb{L}} X_i = 0, \quad X_i^2 = I.$$

There is a deterministic polynomial-time procedure that, given (X_0, X_1) , returns only verified Module-LIP solutions or returns FAIL. If for some promised solution $C_{\text{hid}} : P_0 \rightarrow P_1$ and some $\varepsilon \in \{\pm 1\}$ one has

$$X_1 = \varepsilon C_{\text{hid}} X_0 C_{\text{hid}}^{-1}, \tag{22}$$

then the procedure returns a solution. It uses two rank-one Lenstra–Silverberg computations for each of the two eigenvalue matchings, followed by exact pseudobasis tests.

Proof. Compute the rank-one eigenline modules

$$P_{i,\delta} = P_i \cap \ker(X_i - \delta I) = \mathbf{b}_{i,\delta} v_{i,\delta} \quad (\delta \in \{\pm 1\}).$$

For each tested sign $\varepsilon \in \{\pm 1\}$, apply [theorem 3](#) to the rank-one pairs $P_{0,\delta} \rightarrow P_{1,\varepsilon\delta}$ for $\delta \in \{\pm 1\}$. In the displayed pseudobases, the line map is specified by a scalar $c_\delta \in \mathbb{L}^\times$ satisfying

$$c_\delta \mathbf{b}_{0,\delta} = \mathbf{b}_{1,\varepsilon\delta}, \quad \bar{c}_\delta c_\delta a_{1,\varepsilon\delta} = a_{0,\delta},$$

where $a_{i,\delta} = v_{i,\delta}^* Q_i v_{i,\delta} \in \mathbb{L}^+$. If either rank-one problem has no solution, reject this sign. Otherwise, enumerate the root-of-unity multiples of the two returned

scalars. Each pair defines a unique $\mathbb{L}$ -linear map on the direct sum of the two rational eigenlines. Retain it only if

$$C(P_0) = P_1, \quad C^*Q_1C = Q_0, \quad \det C = 1.$$

The first equality is the integral compatibility test: the module need not equal the direct sum of its two eigenline intersections. Thus every returned map is a verified solution. If [equation \(22\)](#) holds, the two restrictions of C_{hid} occur among the root-of-unity branches for the correct sign, and their assembly passes all three tests. Kernel intersections and module equality are standard HNF/SNF pseudomatrix computations, while $\mu(\mathbb{L})$ is polynomial-time enumerable [13, Lemma 2.5].

Because every assembled map is checked against the original instance, the procedure may safely be run on any finite list of candidate source involutions. It remains to obtain a list that is guaranteed to contain the correct match. The standard cyclotomic reference provides exactly this: its entire shortest shell is explicit.

3.5 The standard cyclotomic shell and the direct SVP reduction

Use the power-of-two cyclotomic coefficient model of [equation \(5\)](#), and put $P = R^2$, $Q = I_2$. In the matrix model define

$$D = \text{diag}(1, -1), \quad E_j = \begin{pmatrix} 0 & \zeta^j \\ \zeta^{-j} & 0 \end{pmatrix} \quad (0 \leq j < N).$$

Proposition 19 (Shape and shortest shell of the standard auxiliary lattice). *The lattice $\mathcal{S}(R^2, I_2)$ has the orthogonal $\mathbb{Z}$ -basis*

$$D, \quad A(s_1, 0), \dots, A(s_{d-1}, 0), \quad E_0, \dots, E_{N-1},$$

with squared lengths

$$1, \quad \underbrace{2, \dots, 2}_{d-1}, \quad \underbrace{1, \dots, 1}_N.$$

Its complete shortest shell is

$$\text{Min}(\mathcal{S}(R^2, I_2)) = \{\pm D\} \cup \{\pm E_j : 0 \leq j < N\}, \quad (23)$$

and the ratio to the next distinct length is $\sqrt{2}$. Every vector in this shell is a self-adjoint involution.

Proof. The coordinate description [equation \(21\)](#) becomes $q(a, z) = a^2 + z\bar{z}$ with $a \in S$ and $z \in R$. The coefficient model makes $1, s_1, \dots, s_{d-1}$ orthogonal with squared lengths $1, 2, \dots, 2$, and makes $1, \zeta, \dots, \zeta^{N-1}$ orthonormal. The displayed basis and shell follow. The involution assertion is also immediate from [equation \(21\)](#).

Theorem 20 (Recovery from one auxiliary SVP solution). *Assume the normalized input is*

$$(R^2, I_2), \quad (R^2, Q),$$

*and is promised to admit a hidden $B \in \text{SL}_2(R)$ with $B^*QB = I_2$. For every $1 \leq \gamma < \sqrt{2}$, one call to γ -SVP on the rank-3d lattice $\mathcal{S}(R^2, Q)$, followed by deterministic polynomial-time exact computations, recovers a verified transition $C \in \text{SL}_2(R)$ satisfying $C^*QC = I_2$.*

Proof. By [proposition 15](#), the hidden map transports the standard auxiliary lattice and its complete shortest shell to the target. The shell gap in [proposition 19](#) forces every γ -SVP output with $\gamma < \sqrt{2}$ to be a target shortest vector. Convert it through Φ to an involution X_1 and enumerate the source shell [equation \(23\)](#). For each source involution X_0 , invoke [proposition 18](#) on (X_0, X_1) . The preimage of X_1 under the hidden conjugation occurs in the shell, so one invocation uses a matched pair and recovers a valid transition. Every output is checked against the defining Module-LIP equations.

This closes the direct reduction from Module-LIP to auxiliary SVP. The remaining subsection produces the same matched target-involution datum in a calibrated lattice, using SVP calls of essentially half the rank.

3.6 Half-rank realization via a calibrated primal–dual lattice

For any normalized rank-two instance define

$$\mathcal{H}(P, Q) = \{A \in \text{End}_R(P) : A^{\dagger Q} = A\},$$

$$\mathcal{T}(P, Q) = \{2A - \text{tr}_{\mathbb{L}}(A)I_2 : A \in \mathcal{H}(P, Q)\}, \quad \Gamma(P, Q) = 2\mathcal{T}(P, Q)^\vee,$$

where the dual is taken in the trace-zero self-adjoint space with the metric of [equation \(16\)](#). The role of Γ is metric calibration rather than a second descent: its primal and dual shortest shells decode to the two types E_j and D already present in [equation \(23\)](#).

Proposition 21 (Calibrated-lattice transport). *The lattices $\mathcal{T}(P, Q)$ and $\Gamma(P, Q)$ are public $\mathbb{Z}$ -lattices of rank $3d$. A normalized isometry C transports both lattices and their metrics.*

Proof. The public endomorphism and adjoint equations compute $\mathcal{H}(P, Q)$. Trace removal has the scalar self-adjoint endomorphisms as kernel and hence has rank $3d$. Conjugation by C preserves integrality, self-adjointness, trace, and the trace metric, and therefore transports $\mathcal{T}$ and its scaled dual.

Proposition 22 (Cyclotomic calibration). *For $P = R^2$ and $Q = I_2$, the lattice Γ has the orthogonal basis*

$$E_0, \dots, E_{N-1}, \quad A(s_1, 0), \dots, A(s_{d-1}, 0), \quad 2D$$

Algorithm 1 Rank-two recovery with half-rank SVP oracles

Require: A normalized target (R^2, Q) promised isometric to (R^2, I_2)
Ensure: A verified $C \in \text{SL}_2(R)$ with $C^*QC = I_2$

- 1: Compute $\Gamma(R^2, Q)$ and run the Bambury–Nguyen procedure of [corollary 23](#).
- 2: **if** a primal shortest vector X_1 is returned **then**
- 3: Set $\mathcal{X}_0 \leftarrow \{\pm E_j : 0 \leq j < N\}$.
- 4: **else**
- 5: Set $X_1 \leftarrow 2Y$ and $\mathcal{X}_0 \leftarrow \{\pm D\}$.
- 6: Verify $X_1^{\dagger Q} = X_1$, $\text{tr}_L X_1 = 0$, and $X_1^2 = I$.
- 7: **for** each $X_0 \in \mathcal{X}_0$ **do**
- 8: Apply [proposition 18](#) to (X_0, X_1) .
- 9: **return** the first verified candidate, if one is found.
- 10: **return** FAIL if no branch passes.

with squared lengths

$$\underbrace{1, \dots, 1}_N, \quad \underbrace{2, \dots, 2}_{d-1}, \quad 4.$$

Consequently

$$\lambda_1(\Gamma) = 1, \quad \lambda_1(\Gamma^\vee) = \frac{1}{2}.$$

The primal shortest vectors are $\pm E_j$, the dual shortest vectors are $\pm D/2$, and both shells have gap $\sqrt{2}$ to the next distinct length.

Proof. The trace-removal lattice has the orthogonal basis

$$D, \quad A(s_1, 0), \dots, A(s_{d-1}, 0), \quad 2E_0, \dots, 2E_{N-1}$$

with squared lengths $1, 2, \dots, 2, 4, \dots, 4$. Dualizing and multiplying by two gives the result.

Corollary 23 (Rank-two recovery with half-rank oracles). *Under the hypotheses of [theorem 20](#), for every fixed $1 \leq \gamma < \sqrt{2}$ a verified transition C is recovered using the Bambury–Nguyen primal–dual algorithm on $\Gamma(R^2, Q)$. Its reduction calls use factor γ , its shell-detection calls are exact, and every SVP call has rank at most*

$$\boxed{\lfloor 3d/2 \rfloor + 1}.$$

Proof. The minima product is $1/2$, and the two relevant shells have gap $\sqrt{2}$. Hence $\gamma^2/2 < 1$ and [theorem 7](#) applies with $m = 3d$. A primal shortest vector is a transported E_j ; twice a dual shortest vector is a transported D . In either case enumerate the corresponding source shell and invoke [proposition 18](#) for each source involution. Thus the Bambury–Nguyen step implements exactly the auxiliary-shortest-vector input required by [theorem 20](#).

Corollary 24 (Two-ended standard-orbit instances). *If both normalized rank-two inputs lie in the special-linear orbit of (R^2, I_2) , recover verified maps C_i from the standard instance to each endpoint using [corollary 23](#). Then $C_1 C_0^{-1}$ solves the two-ended instance. Together with [proposition 4](#), the same conclusion applies to free $\mathrm{GL}_2(R)$ inputs whose normalized branches lie in this standard orbit.*

Proof. The composition $C_1 C_0^{-1}$ maps the first endpoint module to the second and transports its Hermitian form. Its determinant is one. The final statement follows by applying [proposition 4](#) to each endpoint and verifying the resulting candidates in the original instance.

4 Rank four: Hodge descent in $\bigwedge_{\mathbb{L}}^2 V$

We next consider normalized Module-LIP in module rank four. The public endpoints satisfy

$$V_i = \mathbb{L} \otimes_R P_i \simeq \mathbb{L}^4, \quad \mathrm{rank}_{\mathbb{Z}} P_i = 8d, \quad \det Q_i = 1,$$

and the goal is again to recover

$$C(P_0) = P_1, \quad C^* Q_1 C = Q_0, \quad \det C = 1.$$

The role played by the symplectic symmetry in rank two is now played by the Hodge star on $\bigwedge^2 V$. The resulting fixed space has dimension six over $\mathbb{L}^+$.

The logical levels are deliberately separated. The Hodge fixed lattice is constructed for every projective rank-four module. Its metric shape is made explicit when a public pseudobasis is orthogonal. Once a matched map $T = \bigwedge^2 C$ is given, however, the decoder works with an arbitrary public pseudobasis and reduces recovery to four rank-one Lenstra–Silverberg problems. The stronger step from SVP output to such a matched T is proved only for the standard power-of-two cyclotomic reference

$$(R^4, I_4) \longrightarrow (R^4, Q), \quad B \in \mathrm{SL}_4(R), \quad B^* Q B = I_4.$$

For a general orthogonal pseudobasis, the resulting three weighted ideal lattices need not be hypercubic and their shortest vectors need not determine a recognizable basis. We therefore state no full SVP-completion theorem for that broader family.

4.1 The linear Hodge star and the Hermitian involution

We first derive the rank-four symmetry from the wedge product itself. This keeps separate the linear Hodge star, whose matrix is Ω , from the conjugate-linear Hermitian involution that defines the descent.

Recall the ordered exterior basis

$$\mathcal{E}_2 = (e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34}).$$

For an abstract bivector $\xi \in \bigwedge_{\mathbb{L}}^2 V$, write $\mathbf{x} = [\xi]_{\mathcal{E}_2} \in \mathbb{L}^6$ for its coordinate column. In the rest of this section, the base field is unambiguous, so we write $\bigwedge^k V$ for $\bigwedge_{\mathbb{L}}^k V$. With respect to the standard basis (e_1, e_2, e_3, e_4) of V , set

$$\text{vol} := e_1 \wedge e_2 \wedge e_3 \wedge e_4$$

throughout this section. Since $\bigwedge^4 V = \mathbb{L} \text{vol}$, the wedge product defines the $\mathbb{L}$ -bilinear form

$$\beta : \bigwedge^2 V \times \bigwedge^2 V \longrightarrow \mathbb{L}, \quad \xi \wedge \eta = \beta(\xi, \eta) \text{vol}.$$

It is symmetric because two-forms commute:

$$\xi \wedge \eta = (-1)^{2 \cdot 2} \eta \wedge \xi = \eta \wedge \xi.$$

The only nonzero pairings between elements of $\mathcal{E}_2$, up to symmetry, are

$$\beta(e_{12}, e_{34}) = 1, \quad \beta(e_{13}, e_{24}) = -1, \quad \beta(e_{14}, e_{23}) = 1.$$

Indeed, these are the three pairs of complementary index sets; all other pairings between basis bivectors vanish. Hence the Gram matrix of β in $\mathcal{E}_2$ is

$$\Omega = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (24)$$

More precisely, if $\mathbf{x} = [\xi]_{\mathcal{E}_2}$ and $\mathbf{y} = [\eta]_{\mathcal{E}_2}$, then

$$\boxed{\beta(\xi, \eta) = \mathbf{x}^t \Omega \mathbf{y}, \quad \xi \wedge \eta = (\mathbf{x}^t \Omega \mathbf{y}) \text{vol}.} \quad (25)$$

The linear Hodge star $*_0$ for the standard form is most transparent in the self-dual and anti-self-dual basis

$$\begin{aligned} f_1^+ &= e_{12} + e_{34}, & f_2^+ &= e_{13} - e_{24}, & f_3^+ &= e_{14} + e_{23}, \\ f_1^- &= e_{12} - e_{34}, & f_2^- &= e_{13} + e_{24}, & f_3^- &= e_{14} - e_{23}. \end{aligned}$$

It is the diagonal operator

$$*_0 f_i^+ = f_i^+, \quad *_0 f_i^- = -f_i^- \quad (1 \leq i \leq 3).$$

In the ordered basis $\mathcal{E}_2$, one has

$$\begin{aligned} *_0 e_{12} &= e_{34}, & *_0 e_{13} &= -e_{24}, & *_0 e_{14} &= e_{23}, \\ *_0 e_{23} &= e_{14}, & *_0 e_{24} &= -e_{13}, & *_0 e_{34} &= e_{12}. \end{aligned}$$

Thus $[*_0]_{\mathcal{E}_2} = \Omega$. Let $c : \bigwedge^2 V \rightarrow \bigwedge^2 V$ be the coordinatewise CM conjugation determined by

$$[c(\xi)]_{\mathcal{E}_2} = \bar{\mathbf{x}}.$$

We write $*_0$ for the linear Hodge operator and $\mathcal{J}$ for the conjugate-linear Hermitian operator used in the descent; superscript $*$ continues to denote conjugate transpose. Thus

$$\mathcal{J}_0 := *_0 \circ c, \quad \boxed{[\mathcal{J}_0(\xi)]_{\mathcal{E}_2} = \Omega \bar{\mathbf{x}}.} \quad (26)$$

Thus Ω is the matrix of the *linear* Hodge star $*_0$. It is not a linear matrix for $\mathcal{J}_0$; the latter is the semilinear coordinate rule $\mathbf{x} \mapsto \Omega \bar{\mathbf{x}}$.

For $M \in M_4(\mathbb{L})$, put $A_M = C_2(M) = [\bigwedge^2 M]_{\mathcal{E}_2}$. Thus

$$[(\bigwedge^2 M)\xi]_{\mathcal{E}_2} = A_M \mathbf{x}, \quad [(\bigwedge^2 M)\eta]_{\mathcal{E}_2} = A_M \mathbf{y}.$$

The exterior-power identities give

$$\begin{aligned} ((\bigwedge^2 M)\xi) \wedge ((\bigwedge^2 M)\eta) &= (\bigwedge^4 M)(\xi \wedge \eta) \\ &= (\det M)(\xi \wedge \eta). \end{aligned}$$

Using [equation \(25\)](#), the coefficients of vol on the two sides are, respectively,

$$\mathbf{x}^t A_M^t \Omega A_M \mathbf{y} \quad \text{and} \quad (\det M) \mathbf{x}^t \Omega \mathbf{y}.$$

Hence

$$\mathbf{x}^t A_M^t \Omega A_M \mathbf{y} = (\det M) \mathbf{x}^t \Omega \mathbf{y} \quad (\mathbf{x}, \mathbf{y} \in \mathbb{L}^6).$$

Since this holds for all coordinate columns, the rank-four analogue of [equation \(6\)](#) is

$$\boxed{C_2(M)^t \Omega C_2(M) = (\det M) \Omega.} \quad (27)$$

Let $Q_2 = C_2(Q)$ be the Hermitian matrix induced by Q on $\bigwedge^2 V$. For $\mathbf{x} = [\xi]_{\mathcal{E}_2}$ and $\mathbf{y} = [\eta]_{\mathcal{E}_2}$,

$$h_{Q,2}(\xi, \eta) = \bar{\mathbf{x}}^t Q_2 \mathbf{y}.$$

Since [equation \(24\)](#) gives $\Omega^2 = I_6$, the form β is nondegenerate. Hence there is a unique conjugate-linear Hermitian Hodge operator $\mathcal{J}_Q$ characterized intrinsically by

$$\beta(\mathcal{J}_Q \xi, \eta) = h_{Q,2}(\xi, \eta) \quad (\xi, \eta \in \bigwedge^2 V). \quad (28)$$

Indeed, if $\mathbf{j} = [\mathcal{J}_Q(\xi)]_{\mathcal{E}_2}$, then $\mathbf{j}^t \Omega \mathbf{y} = \bar{\mathbf{x}}^t Q_2 \mathbf{y}$ for every $\mathbf{y}$. Using $\Omega^t = \Omega = \Omega^{-1}$ and $Q_2^t = \overline{Q_2}$ gives

$$\boxed{[\mathcal{J}_Q(\xi)]_{\mathcal{E}_2} = \Omega \overline{Q_2} \bar{\mathbf{x}}.} \quad (29)$$

Proposition 25 (Public conjugate of the standard Hermitian Hodge involution). *The map $\mathcal{J}_Q$ is a conjugate-linear involution. If $B \in \mathrm{SL}_4(\mathbb{L})$ is a normalized transition from the standard form to Q , namely $B^*QB = I_4$, put $B_2 = \bigwedge^2 B$ and $A = [B_2]_{\mathcal{E}_2} = C_2(B)$. Then*

$$\boxed{\mathcal{J}_Q = B_2 \circ \mathcal{J}_0 \circ B_2^{-1}}. \quad (30)$$

More generally, if $C^*Q_1C = Q_0$ and $\det C = 1$, then

$$\left(\bigwedge^2 C\right) \circ \mathcal{J}_{Q_0} = \mathcal{J}_{Q_1} \circ \left(\bigwedge^2 C\right). \quad (31)$$

Proof. Since $Q_2 = Q_2^*$ and $\det Q = 1$, applying equation (27) to Q gives $\overline{Q_2}\Omega Q_2 = \Omega$, hence $\Omega\overline{Q_2}\Omega Q_2 = I$. Therefore equation (29) gives

$$[\mathcal{J}_Q^2(\xi)]_{\mathcal{E}_2} = \Omega\overline{Q_2}\Omega Q_2 \mathbf{x} = \mathbf{x},$$

so $\mathcal{J}_Q^2 = I$. If $B^*QB = I_4$, then $Q_2 = (A^{-1})^*A^{-1}$ and $\overline{Q_2} = A^{-t}\overline{A}^{-1}$. Applying equation (27) to B^t gives $A\Omega A^t = \Omega$, or equivalently $\Omega A^{-t} = A\Omega$. Therefore

$$[\mathcal{J}_Q(\xi)]_{\mathcal{E}_2} = \Omega A^{-t}\overline{A}^{-1} \overline{\mathbf{x}} = A\Omega\overline{A}^{-1} \overline{\mathbf{x}} = [(B_2 \circ \mathcal{J}_0 \circ B_2^{-1})(\xi)]_{\mathcal{E}_2}.$$

The general transport equation follows by the same calculation from $Q_{0,2} = C_2(C)^*Q_{1,2}C_2(C)$ and $C_2(C)^t\Omega C_2(C) = \Omega$.

If $D = B^{-1}$ and $Q = D^*D$, the same identity reads

$$\mathcal{J}_Q = \left(\bigwedge^2 D\right)^{-1} \circ \mathcal{J}_0 \circ \bigwedge^2 D.$$

Thus the rank-two and rank-four formulas have the same conjugation form.

Definition 26 (Rank-four fixed space).

$$W_4(Q) = \mathrm{Fix}(\mathcal{J}_Q) \subseteq \bigwedge^2 V.$$

Proposition 27 (Hodge fixed space). *The space $W_4(Q)$ has dimension six over $\mathbb{L}^+$, and*

$$\mathbb{L} \otimes_{\mathbb{L}^+} W_4(Q) \simeq \bigwedge_{\mathbb{L}}^2 V.$$

If $B \in \mathrm{SL}_4(\mathbb{L})$ satisfies $B^*QB = I_4$, then

$$W_4(Q) = \left(\bigwedge^2 B\right)(W_4(I_4)).$$

Proof. Apply lemma 6 to $\mathcal{J}_Q$ and use equation (30).

4.2 The integral Hodge lattice, its metric, and its shape

The Hodge fixed space is the rank-four analogue of $W_2(Q)$. We now turn it into a public Euclidean lattice and then compute its coordinates under the additional assumption of an orthogonal pseudobasis.

For $\xi, \eta \in W_4(Q)$, the defining identity [equation \(28\)](#) and $\mathcal{J}_Q \xi = \xi$ give

$$h_{Q,2}(\xi, \eta) = \beta(\xi, \eta).$$

The right-hand side lies in $\mathbb{L}^+$: Hermitian symmetry gives $\overline{h_{Q,2}(\xi, \eta)} = h_{Q,2}(\eta, \xi)$, while β is symmetric. Thus the restriction of the induced Hermitian form is a positive symmetric $\mathbb{L}^+$ -bilinear form. Put

$$q_{Q,2}(\xi) = \frac{1}{2} h_{Q,2}(\xi, \xi) = \frac{1}{2} \bar{\mathbf{x}}^t Q_2 \mathbf{x}, \quad \mathbf{x} = [\xi]_{\mathcal{E}_2}, \quad (32)$$

and define

$$\langle \xi, \eta \rangle_4 = \frac{1}{2[\mathbb{L} : \mathbb{Q}]} \operatorname{Tr}_{\mathbb{L}/\mathbb{Q}} h_{Q,2}(\xi, \eta) = \frac{1}{d} \operatorname{Tr}_{\mathbb{L}^+/\mathbb{Q}} \frac{h_{Q,2}(\xi, \eta)}{2}. \quad (33)$$

Again this is one half of the normalized trace of the ambient exterior-square metric.

Definition 28 (Rank-four auxiliary lattice).

$$\mathcal{W}(P, Q) = W_4(Q) \cap \bigwedge_R^2 P, \quad (34)$$

with the metric [equation \(33\)](#).

Theorem 29 (Integral transport). *The lattice $\mathcal{W}(P, Q)$ is a full S -lattice of rank six and hence has $\mathbb{Z}$ -rank $6d$. If*

$$C(P_0) = P_1, \quad C^* Q_1 C = Q_0, \quad \det C = 1,$$

then

$$\bigwedge^2 C : \mathcal{W}(P_0, Q_0) \xrightarrow{\sim} \mathcal{W}(P_1, Q_1)$$

is an S -linear Euclidean isometry.

Proof. In the exact public representation, $\mathcal{J}_Q$ is $\mathbb{Q}$ -linear on the underlying rational space, so its fixed space is a rational subspace of dimension $6d$. Its intersection with the full lattice $\bigwedge_R^2 P$ is therefore a full $\mathbb{Z}$ -lattice in that fixed space. It is stable under S and has S -rank six. The operator transport is [equation \(31\)](#). The equations $C(P_0) = P_1$ and $C^* Q_1 C = Q_0$ directly show that $\bigwedge^2 C$ preserves the integral exterior-power module and its induced Hermitian form.

Remark 30 (Metric and coefficient geometry). The ambient exterior metric is the canonical one and is preserved by every hidden transition. The factor $1/2$ is a common scaling and therefore does not affect shortest vectors or approximation factors. It is chosen because a fixed bivector contains three coordinates together with their conjugate complements; after division by two, the standard fixed-space coordinates have exactly the coefficient norm. This is the same normalization principle as in [remark 13](#).

Proposition 31 (Diagonal Hodge-lattice coordinates). *Assume that*

$$P = \bigoplus_{i=1}^4 \mathbf{a}_i e_i, \quad Q = \text{diag}(\alpha_1, \alpha_2, \alpha_3, \alpha_4), \quad \alpha_1 \alpha_2 \alpha_3 \alpha_4 = 1$$

is an orthogonal pseudobasis. Every fixed bivector is uniquely

$$\begin{aligned} w(a, b, c) = & ae_{12} + be_{13} + ce_{14} \\ & + \alpha_1 \alpha_4 \bar{c} e_{23} - \alpha_1 \alpha_3 \bar{b} e_{24} + \alpha_1 \alpha_2 \bar{a} e_{34}. \end{aligned} \quad (35)$$

The parameters lie in

$$\begin{aligned} \mathfrak{A}_{12} &= \mathbf{a}_1 \mathbf{a}_2 \cap \frac{1}{\alpha_1 \alpha_2} \bar{\mathbf{a}}_3 \bar{\mathbf{a}}_4, \\ \mathfrak{A}_{13} &= \mathbf{a}_1 \mathbf{a}_3 \cap \frac{1}{\alpha_1 \alpha_3} \bar{\mathbf{a}}_2 \bar{\mathbf{a}}_4, \quad \mathfrak{A}_{14} = \mathbf{a}_1 \mathbf{a}_4 \cap \frac{1}{\alpha_1 \alpha_4} \bar{\mathbf{a}}_2 \bar{\mathbf{a}}_3. \end{aligned}$$

The quadratic form is

$$q_{Q,2}(w(a, b, c)) = \alpha_1 \alpha_2 a \bar{a} + \alpha_1 \alpha_3 b \bar{b} + \alpha_1 \alpha_4 c \bar{c}. \quad (36)$$

Hence

$$\mathcal{W}(P, Q) \simeq (\mathfrak{A}_{12}, \alpha_1 \alpha_2) \perp (\mathfrak{A}_{13}, \alpha_1 \alpha_3) \perp (\mathfrak{A}_{14}, \alpha_1 \alpha_4)$$

as an orthogonal sum of rank-one Hermitian ideal lattices with the indicated weights.

Proof. For diagonal Q , the matrix Q_2 is diagonal in the exterior basis, with entries $\alpha_i \alpha_j$. Solving the coordinate fixed-point equation

$$\mathbf{x} = \Omega \overline{Q_2} \bar{\mathbf{x}}, \quad \mathbf{x} = [\xi]_{\mathcal{E}_2},$$

pairs each coordinate with its complementary index and gives exactly [equation \(35\)](#); the determinant-one equation is used to rewrite, for example, $x_{12} = \alpha_3 \alpha_4 \bar{x}_{34}$ as $x_{34} = \alpha_1 \alpha_2 \bar{x}_{12}$. Intersecting these pairs with [equation \(4\)](#) yields the three displayed ideal intersections. Finally, substituting [equation \(35\)](#) into $q_{Q,2} = h_{Q,2}/2$ gives [equation \(36\)](#); its polarization has no mixed terms between a, b, c .

For $P = R^4$ and $Q = I_4$ this becomes

$$[w(a, b, c)]_{\mathcal{E}_2} = (a, b, c, \bar{c}, -\bar{b}, \bar{a})^t, \quad q(a, b, c) = a \bar{a} + b \bar{b} + c \bar{c}. \quad (37)$$

In the power-of-two cyclotomic coefficient model, define

$$u_{1,j} = w(\zeta^j, 0, 0), \quad u_{2,j} = w(0, \zeta^j, 0), \quad u_{3,j} = w(0, 0, \zeta^j) \quad (0 \leq j < N).$$

Their union $\mathcal{B}_0$ is an orthonormal $\mathbb{Z}$ -basis of $\mathcal{W}(R^4, I_4)$. Thus the standard Hodge lattice is hypercubic of rank $3N = 6d$, its shortest shell is $\{\pm u_{k,j}\}$, and its next distinct length is $\sqrt{2}$.

4.3 Recovering the transition from a matched exterior-square map

The fixed Hodge lattice lowers the dimension, but this is useful only if a candidate map on the exterior square can be converted back to the original four-dimensional transition. We avoid any classification of exterior-square isometries. Instead, assuming that the correct branch is $T = \bigwedge^2 C$, we recover the four image lines directly and then solve four rank-one module-isomorphism problems. The resulting decoder works for an arbitrary public pseudobasis of the reference module.

For a four-dimensional $\mathbb{L}$ -space W and a nonzero vector $v \in W$, put

$$\mathcal{U}_W(v) = v \wedge W \subseteq \bigwedge^2 W$$

and, for a subspace $U \subseteq \bigwedge^2 W$, put

$$\text{Ann}_\wedge(U) = \{x \in W : x \wedge u = 0 \text{ for every } u \in U\}.$$

Lemma 32 (Recovering a line from $v \wedge W$). *For every nonzero $v \in W$,*

$$\boxed{\text{Ann}_\wedge(\mathcal{U}_W(v)) = \mathbb{L}v.}$$

Moreover, for every isomorphism $C : W_0 \rightarrow W_1$,

$$\left(\bigwedge^2 C\right)\mathcal{U}_{W_0}(v) = \mathcal{U}_{W_1}(Cv).$$

Both spaces are computed by linear systems over $\mathbb{L}$.

Proof. Every scalar multiple of v annihilates $v \wedge W$. Conversely, if $x \notin \mathbb{L}v$, choose $w \notin \text{span}_{\mathbb{L}}(x, v)$. Then $x \wedge v \wedge w \neq 0$, so x is not in the annihilator. The transport identity follows from $(\bigwedge^2 C)(v \wedge w) = Cv \wedge Cw$ and the surjectivity of C .

Proposition 33 (Four-line Lenstra–Silverberg decoder). *Let (P_i, Q_i) be a normalized rank-four Module-LIP instance, and fix any pseudobasis of the reference module*

$$P_0 = \bigoplus_{i=1}^4 \mathfrak{a}_i e_i.$$

Put

$$\alpha_i = h_{Q_0}(e_i, e_i) \in \mathbb{L}^+ \quad (1 \leq i \leq 4).$$

There is a deterministic polynomial-time procedure that, given an $\mathbb{L}$ -linear isomorphism

$$T : \bigwedge_{\mathbb{L}}^2 V_0 \longrightarrow \bigwedge_{\mathbb{L}}^2 V_1,$$

returns only verified Module-LIP solutions or returns FAIL. If

$$T = \bigwedge^2 C_{\text{hid}}$$

for some promised solution $C_{\text{hid}} : P_0 \rightarrow P_1$, then the procedure returns a solution. It uses four rank-one Lenstra–Silverberg computations, root-of-unity enumeration, and exact module arithmetic.

Proof. For $1 \leq i \leq 4$, compute

$$\ell_i = \text{Ann}_{\wedge}(T(\mathcal{U}_{V_0}(e_i))).$$

Reject unless every ℓ_i is an $\mathbb{L}$ -line. Intersect it with the target module and compute a rank-one pseudobasis

$$P_{1,i} = P_1 \cap \ell_i = \mathfrak{b}_i v_i, \quad q_i = v_i^* Q_1 v_i \in \mathbb{L}^+.$$

For each i , apply [theorem 3](#) to the rank-one Hermitian lattices

$$(\mathfrak{a}_i, \alpha_i) \longrightarrow (\mathfrak{b}_i, q_i).$$

If one of these problems has no solution, return FAIL. Otherwise recover a scalar c_i satisfying

$$c_i \mathfrak{a}_i = \mathfrak{b}_i, \quad q_i c_i \bar{c}_i = \alpha_i,$$

and enumerate its root-of-unity multiples. Since the module rank is fixed at four and $\mu(\mathbb{L})$ is polynomial-time enumerable, the total number of four-tuples remains polynomial. For every four-tuple of branches, define C by $C(e_i) = c_i v_i$ and retain it only if

$$C(P_0) = P_1, \quad C^* Q_1 C = Q_0, \quad \det C = 1, \quad \bigwedge^2 C = T. \quad (38)$$

These tests ensure that every returned matrix is a verified solution and that it is compatible with the candidate auxiliary map.

It remains to prove completeness for a matched T . If $T = \bigwedge^2 C_{\text{hid}}$, then [lemma 32](#) gives

$$\ell_i = \mathbb{L} C_{\text{hid}} e_i.$$

Moreover,

$$P_1 \cap \ell_i = C_{\text{hid}}(\mathfrak{a}_i e_i), \quad P_1 = \bigoplus_{i=1}^4 (P_1 \cap \ell_i). \quad (39)$$

Indeed, applying C_{hid}^{-1} to an element of $P_1 \cap \mathbb{L}C_{\text{hid}}e_i$ places it in $P_0 \cap \mathbb{L}e_i = \mathfrak{a}_i e_i$, and the direct-sum identity is the image of the source pseudobasis decomposition. Hence the four restrictions of C_{hid} occur among the root-of-unity branches of the Lenstra–Silverberg outputs. Their assembly passes [equation \(38\)](#). All line intersections and module equalities are exact pseudobasis computations. Orthogonality is not used in this argument: the four rank-one calls recover the line restrictions, while the final equation $C^*Q_1C = Q_0$ enforces all cross terms.

Boundary between the general decoder and the SVP step. [Proposition 33](#) applies to every projective reference module once a pseudobasis and the matched exterior-square map are given. The additional orthogonality assumption in [proposition 31](#) is used only to write the Hodge lattice as three weighted ideal summands. Neither result by itself produces the matched map from SVP. For a general orthogonal pseudobasis, those weighted ideal summands need not be hypercubic, and a shortest vector need not extend to a publicly recognizable shortest basis. The next subsection proves basis recovery and completion only for the standard power-of-two cyclotomic reference. Extending that step to broader weighted ideal families is left as future work.

4.4 Obtaining a matched exterior-square map in the standard cyclotomic orbit

For the standard lattice in [equation \(37\)](#), a single shortest vector determines only one direction in the six-dimensional fixed space and therefore cannot determine $\bigwedge^2 B$. We first recover a complete orthonormal shortest basis and then use the public real-subfield action to match it with the standard basis.

Lemma 34 (Recovering a hypercubic basis by successive SVP). *Let Λ be a rank- m hypercubic lattice with shortest length one and second distinct length $\sqrt{2}$. For every $1 \leq \gamma < \sqrt{2}$, an orthonormal $\mathbb{Z}$ -basis of Λ is recovered with at most m γ -SVP calls, in ranks $m, m-1, \dots, 1$.*

Proof. In coordinates with respect to an orthonormal lattice basis, every squared length is a sum of integer squares. Hence a γ -SVP output with $\gamma < \sqrt{2}$ lies in the shortest shell and is, up to sign, one of those basis vectors. After recovering $x_1, \dots, x_k$, compute exactly

$$\Lambda_k = \Lambda \cap \text{span}_{\mathbb{R}}(x_1, \dots, x_k)^\perp.$$

It is the hypercubic lattice spanned by the remaining basis vectors. Iterate.

Assume now that $N = [\mathbb{L} : \mathbb{Q}] \geq 4$ is a power of two and put $s = \zeta + \zeta^{-1}$, so $S = \mathbb{Z}[s]$. Multiplication by s acts on the standard basis as

$$su_{k,j} = u_{k,j-1} + u_{k,j+1}, \tag{40}$$

where $u_{k,-1} = -u_{k,N-1}$ and $u_{k,N} = -u_{k,0}$. Its support graph is the disjoint union of three signed N -cycles, each with edge-sign product -1 .

Proposition 35 (Signed-cycle completion). *Let (R^4, Q) lie in the normalized special-linear orbit of (R^4, I_4) , and let $\mathcal{B}_1$ be any orthonormal $\mathbb{Z}$ -basis of $\mathcal{W}(R^4, Q)$. From the public action of s one can enumerate at most*

$$6(4N)^3$$

signed bijections $T_W : \mathcal{B}_0 \rightarrow \mathcal{B}_1$ whose $\mathbb{Z}$ -linear extensions commute with s . Every such map extends S -linearly on the Hodge lattices, and the restriction

$$\left(\bigwedge^2 B\right)|_{W_4(I_4)}$$

of the hidden exterior-square map occurs in the list.

Proof. The hidden images of $\mathcal{B}_0$ form an orthonormal integral basis of the target lattice. Two orthonormal $\mathbb{Z}$ -bases differ by a signed permutation. In the hidden basis, multiplication by s is the union of the three signed cycles in [equation \(40\)](#); conjugating by a signed permutation preserves this signed support graph. There are six component matchings. Between two components there are $2N$ underlying cycle isomorphisms. Once the sign of one image vertex is chosen, the intertwining equations determine all remaining signs; consistency around the cycle holds because both edge-sign products are -1 . Hence there are at most $4N$ signed intertwiners per component and $6(4N)^3$ in total. Since $S = \mathbb{Z}[s]$, commuting with s is exactly S -linearity. The hidden map is one of the enumerated intertwiners.

Each T_W extends first $\mathbb{L}^+$ -linearly from the full S -lattice to $W_4(I_4)$ and then uniquely $\mathbb{L}$ -linearly to

$$\bigwedge_{\mathbb{L}}^2 (\mathbb{L}^4)$$

by [lemma 6](#). We denote the latter extension by T and test it by the exact decoder of [proposition 33](#).

4.5 The direct SVP-to-Module-LIP reduction

Theorem 36 (Rank-four recovery from the Hodge-lattice SVP). *Assume $N = [\mathbb{L} : \mathbb{Q}] \geq 4$ is a power of two and the normalized input is*

$$(R^4, I_4), \quad (R^4, Q),$$

*promised to admit a hidden $B \in \mathrm{SL}_4(R)$ with $B^*QB = I_4$. For every $1 \leq \gamma < \sqrt{2}$, a verified transition $C \in \mathrm{SL}_4(R)$ satisfying $C^*QC = I_4$ is recovered using at most $6d$ calls to γ -SVP in ranks at most $6d$, followed by a polynomial-size enumeration and exact number-field computations.*

Proof. The map $\bigwedge^2 B$ sends the standard Hodge lattice to $\mathcal{W}(R^4, Q)$. First use [lemma 34](#) to recover an orthonormal basis $\mathcal{B}_1$. Next use [proposition 35](#) to enumerate a polynomial list containing the restriction of $\bigwedge^2 B$. Extend each candidate to the full exterior square and run the decoder of [proposition 33](#) with the standard pseudobasis. The exact module, Hermitian, determinant, and compound-matrix tests accept the hidden branch and reject every invalid branch.

Algorithm 2 Rank-four recovery with half-rank SVP oracles

Require: A normalized target (R^4, Q) promised isometric to (R^4, I_4)
Ensure: A verified $C \in \text{SL}_4(R)$ with $C^*QC = I_4$
 1: Compute $\mathcal{W}(R^4, Q)$ and recover an orthonormal basis $\mathcal{B}_1$ by [corollary 37](#).
 2: Compute the public action of $s = \zeta + \zeta^{-1}$ in $\mathcal{B}_1$.
 3: **for** each signed-cycle intertwiner T_W from [proposition 35](#) **do**
 4: Extend T_W to an $\mathbb{L}$ -linear map T on $\bigwedge_{\mathbb{L}}^2(\mathbb{L}^4)$.
 5: Apply [proposition 33](#) to the standard pseudobasis and T .
 6: **return** the first verified transition, if one is found.
 7: **return** FAIL if no branch passes.

This closes the direct reduction from the original rank-four Module-LIP instance to shortest-vector computations on the constructed Hodge lattice. The next subsection changes only the ranks of those SVP calls.

4.6 Half-rank realization for the hypercubic Hodge lattice

Corollary 37 (Bambury–Nguyen basis recovery). *For the standard cyclotomic Hodge lattice of rank $6d$, the Bambury–Nguyen hypercubic algorithm recovers an orthonormal basis using γ -SVP calls in ranks at most*

$$\boxed{3d + 1}$$

for every

$$1 \leq \gamma < \sqrt{2 - \frac{2}{6d}}.$$

Proof. Apply the hypercubic clause of [theorem 7](#) with $m = 6d$.

Theorem 38 (Rank-four recovery with half-rank oracles). *Under the hypotheses of [theorem 36](#), for every $1 \leq \gamma < \sqrt{2 - 2/(6d)}$ a verified transition C is recovered with SVP oracle calls only in ranks at most $3d + 1$, together with polynomial-time exact computations.*

Proof. Use [corollary 37](#) in place of the successive full-rank basis recovery in [theorem 36](#). The signed-cycle completion and the four-line Lenstra–Silverberg decoder are unchanged.

Corollary 39 (Two-ended standard-orbit instances). *If both normalized rank-four inputs lie in the special-linear orbit of (R^4, I_4) , recover verified maps C_i from the standard instance to each endpoint using [theorem 38](#). Then $C_1 C_0^{-1}$ solves the two-ended instance. Together with [proposition 4](#), the same conclusion applies to free $\text{GL}_4(R)$ inputs whose normalized branches lie in this standard orbit.*

Proof. As in [corollary 24](#), the composition transports the module and Hermitian form from the first endpoint to the second and has determinant one. Determinant normalization and final verification handle the stated free $\text{GL}_4(R)$ instances.

5 Scope of the reductions

The constructions, decoders, and complete attacks hold under different hypotheses. We summarize the precise boundary of each statement to prevent the lower-dimensional fixed-space construction from being mistaken for a full search reduction.

1. *General normalized descent.* For arbitrary projective rank-two and rank-four modules, the public R -action and trace Gram matrix determine the Hermitian form, the conjugate-linear symmetry, and the fixed auxiliary lattice. The transport results [proposition 15](#) and [theorem 29](#) are unconditional under the normalized determinant-one promise.
2. *Decoding matched auxiliary data.* In rank two, [proposition 18](#) decodes any matched pair of integral self-adjoint involutions and checks the integral module equality directly. In rank four, [proposition 33](#) decodes a matched exterior-square map from any public pseudobasis of the reference module. It uses four rank-one Lenstra–Silverberg computations and does not require either endpoint to be diagonal.
3. *Complete SVP-to-Module-LIP reductions.* [Theorems 20](#) and [36](#) prove this stronger level only for the standard free power-of-two cyclotomic orbit. In rank two the complete source shortest shell is explicit. In rank four the standard Hodge lattice is hypercubic and its public S -action has the signed-cycle form used in [proposition 35](#). The two-ended corollaries cover instances whose two endpoints can each be connected to that standard reference.
4. *What remains beyond the standard rank-four orbit.* [Proposition 31](#) gives the exact orthogonal sum of three weighted ideal lattices whenever the reference pseudobasis is orthogonal, and [proposition 33](#) gives the complete decoder once the correct auxiliary map is known. What is not proved is a polynomial procedure that, from SVP data on an arbitrary such weighted sum, completes a shortest-vector output to a list containing the hidden S -linear Hodge isometry. The lattice may fail to be hypercubic and its shortest vectors may not form a recognizable basis. This basis-completion problem is the remaining obstacle to a full attack on the orthogonal-pseudobasis family.
5. *Normalization and oracle rank.* The determinant-one condition is part of the normalized problem. [Proposition 4](#) handles the stated free $\mathrm{GL}_r(R)$ branches; a general projective determinant line requires separate ideal and norm data. Bambury–Nguyen is invoked only after the direct decoding chains are closed. It reduces the ranks of the SVP calls but does not perform eigenline recovery, Hodge-basis completion, or Lenstra–Silverberg assembly.
6. *Metric choice.* The factor $1/2$ in the two auxiliary metrics is a global scaling of the canonical tensor-induced trace metric. It preserves shortest vectors, approximation factors, and primal–dual minima products, while making the standard fixed-coordinate models equal to the ordinary coefficient lattices used in the shell calculations.

6 Conclusion

We developed linear descents for normalized Module-LIP in module ranks two and four. In both cases, a determinant identity turns the hidden transition into conjugation of a canonical conjugate-linear symmetry. The fixed space of that symmetry is public, has lower dimension over the maximal real subfield, and inherits an intrinsic metric from the corresponding tensor representation.

In rank two, the symmetric-square fixed lattice is naturally the lattice of integral trace-zero self-adjoint endomorphisms. Its shortest vectors are involutions, and a matched involution pair exposes two rank-one module isomorphisms. This yields a complete SVP reduction for the standard cyclotomic orbit; the calibrated primal–dual construction and Bambury–Nguyen reduce the largest oracle rank to $\lfloor 3d/2 \rfloor + 1$.

In rank four, the Hermitian Hodge involution gives a rank- $6d$ fixed lattice. An orthogonal pseudobasis identifies it with three weighted ideal lattices, while a matched exterior-square map can be decoded for an arbitrary public pseudobasis by recovering four image lines and solving four rank-one problems. For the standard cyclotomic orbit, the Hodge lattice is hypercubic and the public real-subfield action completes a recovered shortest basis to a polynomial list containing the hidden exterior-square map. The largest oracle rank is then $3d + 1$.

The remaining rank-four obstacle is now isolated: for general weighted ideal summands, one needs a public method that converts SVP information into a matched S -linear Hodge-lattice isometry. The fixed lattice and the lifting procedure already apply; what is missing is the family-specific basis-completion step.

A Computing the auxiliary lattices from public data

The main text defines the auxiliary lattices intrinsically. For completeness, this appendix explains how to compute their $\mathbb{Z}$ -bases and exact Gram matrices from the public order action and trace Gram matrix. No new reduction is introduced here. We clear denominators in all rational systems and compute their integral solution lattices by Hermite or Smith normal form.

A.1 Rank two

Let $M = 4d$ and let $u_1, \dots, u_M$ be the public $\mathbb{Z}$ -basis of P . Let G be the Gram matrix of g_Q , and let $M_i = \rho(\omega_i)$ be the action matrices of a $\mathbb{Z}$ -basis $\omega_1, \dots, \omega_{2d}$ of R . Write an unknown $\mathbb{Z}$ -endomorphism of P as an integer matrix $X \in M_M(\mathbb{Z})$. The equations

$$XM_i = M_iX, \quad X^t G = GX, \quad \text{tr}_{\mathbb{Q}}(M_i X) = 0 \quad (1 \leq i \leq 2d) \quad (41)$$

impose, respectively, R -linearity, self-adjointness, and $\mathbb{L}$ -trace zero. The last assertion follows from

$$\text{tr}_{\mathbb{Q}}(\rho(a)X) = \text{Tr}_{\mathbb{L}/\mathbb{Q}}(a \text{tr}_{\mathbb{L}}(X))$$

and the nondegeneracy of the field-trace pairing. Hence the integer kernel of [equation \(41\)](#) is exactly

$$\Phi(\mathcal{S}(P, Q)) = \{X \in \text{End}_R(P) : X^{\dagger_Q} = X, \text{tr}_{\mathbb{L}} X = 0\}.$$

When tensor coordinates are needed, reconstruct the corresponding 2-by-2 $\mathbb{L}$ -matrix and apply Φ^{-1} .

The same commutation and adjoint equations, without the trace equations, compute $\mathcal{H}(P, Q)$. Applying the trace-removal map gives $\mathcal{T}(P, Q)$, and exact duality with its rational Gram matrix gives $\Gamma(P, Q)$. For an involution X , rational kernel computation followed by intersection with P gives

$$P_{\pm} = P \cap \ker(X \mp I) = \mathfrak{b}_{\pm} v_{\pm}.$$

After the two rank-one maps are assembled, the equality $C(P_0) = P_1$ is checked directly from the original pseudobases. No direct-sum decomposition of either integral module is assumed.

A.2 Rank four

First compute a pseudobasis, and hence a $\mathbb{Z}$ -basis, of $\bigwedge_R^2 P$ from [equation \(4\)](#). Reconstruct Q by [theorem 1](#), form $Q_2 = C_2(Q)$, and use [equation \(29\)](#) together with the public matrix of CM conjugation to express $\mathcal{J}_Q$ as a rational linear map on

$$\left(\bigwedge_R^2 P\right) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq \bigwedge_{\mathbb{L}}^2 V.$$

The fixed rational space is the kernel of the rational matrix of $\mathcal{J}_Q - I$; in a $\mathbb{Q}$ -basis of the exterior-power module this is a $12d$ -by- $12d$ rational matrix. Intersecting that kernel with $\bigwedge_R^2 P$ gives $\mathcal{W}(P, Q)$. Restricting the public R -action to S gives the public S -action on this lattice, and [equation \(33\)](#) gives its exact rational Gram matrix.

Extending a candidate fixed-lattice map. An S -linear map T_W between two fixed lattices extends first to an $\mathbb{L}^+$ -linear map between the fixed spaces. Choose $\vartheta \in \mathbb{L}$ with $\bar{\vartheta} = -\vartheta$. By [lemma 6](#), every exterior-square vector is uniquely $w_0 + \vartheta w_1$ with w_0, w_1 in the fixed space, and the required $\mathbb{L}$ -linear extension is

$$T(w_0 + \vartheta w_1) = T_W(w_0) + \vartheta T_W(w_1).$$

The line spaces in [proposition 33](#) are then obtained by ordinary linear kernels over $\mathbb{L}$, and their intersections with P_1 are computed by the same pseudobasis normal-form routines. These operations are deterministic polynomial-time in the standard number-field and ideal encodings [4,10].

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